

Signal, State, or Sensor Style? Generalizability and Interpretive Limits of Automated Facial-Inference Channels During Cognitive Assessment

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ABSTRACT. Commercial facial-expression software can be attached to a computer-administered cognitive assessment so that, while a student works through the items, the software logs a dense stream of emotion-labeled output. Such channels are increasingly analyzed as measures of emotion during testing, and their volume, often hundreds of thousands of time-stamped records in a modest study, invites the conclusion that they must be measuring something well. That conclusion conflates three properties requiring separate evidence: how much was recorded, whether the resulting scores distinguish one person from another consistently, and whether they add information about outcomes of substantive interest. We separate the three properties in archival data from a 25-item untimed reasoning task in which 169 college students, tested partly in person and partly over Zoom after a COVID-19 interruption, were monitored by a commercial system emitting nine facial-inference channels. Since adjacent observations of the same face are strongly autocorrelated, we evaluate repeatability under deliberately conservative accountings of the information the stream contains and corroborate the model-based results with model-free item splits. Eight of the nine activation channels order persons consistently even under the harshest within-cell accounting; the sparse Joy channel is the instructive exception. The channels also share a common component, but recording volume and shared software processing remain rival explanations for it. Despite this repeatability, the evaluated channel scores improved held-out prediction of no behavioral or survey benchmark. Repeatability and usefulness therefore required separate evidence, and we summarize the workflow as reporting guidance that keeps observation volume, repeatability, and utility distinct.

Keywords: process data; automated facial expression analysis; generalizability theory; reliability; measurement validity; intensive longitudinal data

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1. Introduction

Cognitive assessments administered on computers now record more than item responses. It has become routine to capture the process that surrounds each response, including response times, actions such as hint requests and skips, and, increasingly, continuous sensor streams recorded while the participant works (Chen, 2020; Fu et al., 2024; Tang et al., 2021; Zhang et al., 2023). Among the richest of these streams are the outputs of automated facial-expression analysis. Off-the-shelf systems can process a webcam feed and emit emotion-labeled numerical channels several times per second, promising what self-report cannot provide: a real-time record of affect-relevant behavior during problem solving (Bosch et al., 2016; Calvo & D’Mello, 2010; D’Mello & Graesser, 2012; Monkaresi et al., 2017). Validation studies of commercial classifiers (Dupré et al., 2020; Kulke et al., 2020; Küntzler et al., 2021; Stöckli et al., 2018) and a growing literature on affect during learning (Bian et al., 2018; Horvers et al., 2024; Kuklick et al., 2026; Pekrun et al., 2002) have made such channels easy to adopt. It is much less clear, however, what the resulting data are measurements of, and how much measurement information they contain. The title of this article names the three families of interpretation at stake: a repeatable channel score could reflect a psychological signal, a transient state, or merely a sensor style, by which we mean stable characteristics of how a particular person appears to a particular capture pipeline. In this paper we take one such data stream through the full chain of measurement questions that precedes any substantive use, and we find that the answers come apart in an instructive way.

A first difficulty concerns density itself. A facial-inference stream sampled several times per second across a half-hour session yields hundreds of thousands of rows per study, and this volume is easily mistaken for measurement precision. However, adjacent observations of the same face are strongly serially dependent, so that the number of recorded observations can exceed the number of effectively independent observations by an order of magnitude. This concern is familiar from intensive longitudinal research, where summaries computed from finite, autocorrelated within-person streams can be far less stable than the raw data volume suggests (Castro-Alvarez et al., 2026; Cranford et al., 2006; McNeish, 2026; Schneider & Junghaenel, 2023; van Alphen et al., 2022; Xiao et al., 2026), and where increasingly complex dynamic summaries have been shown to add little criterion information beyond simple ones (Dejonckheere et al., 2019). Dense process channels attached to assessments are subject to both problems.

A second difficulty is that even genuinely repeatable scores need not be useful ones. Reliability is not a fixed property of an instrument but depends on the score considered, the decision to be made, and the universe of admissible observations (Ellis, 2013, 2021; Markon, 2023). Tasks that produce robust group-level effects can simultaneously produce weak individual-difference measures (Hedge et al., 2018), and scores that order persons consistently can still fail to add information about any criterion of interest (Pratiwi et al., 2024; Yarkoni & Westfall, 2017). Generalizability theory provides the natural vocabulary for the repeatability question, since it asks across which facets of observation a person score generalizes, and at what cost in error variance (Brennan, 2001; Cronbach et al., 1972; Shavelson & Webb, 1991; Shavelson et al., 1989; Webb et al., 2006). Out-of-sample prediction against specified baselines provides the natural framing for the utility question. Because the two questions differ in logic, they can receive different answers from the same data.

A third difficulty is specific to facial inference: a construct-validity ceiling constrains the interpretation of everything the software emits. Facial movements do not map one-to-one onto specific subjective emotions; the same configuration can express different states across persons and contexts, and context can dominate the face in emotion inference (Barrett et al., 2019; Goel et al., 2024; Krumhuber et al., 2023). Commercial classifiers disagree with each other and degrade on non-standardized expressions (Dupré et al., 2020; Küntzler et al., 2021), and automated outputs and self-reports can yield discrepant accounts of the same episode (Horvers et al., 2024; Kuklick et al., 2026). Transparent measurement practice therefore requires reporting what the recorded variables are, namely software outputs with documented provenance, rather than renaming them as the constructs they suggest (Borsboom et al., 2004; Campbell & Fiske, 1959; Cronbach & Meehl, 1955; Flake & Fried, 2020). Throughout this article we accordingly treat the nine recorded channels as project-defined variables whose names are labels, not construct claims, and we treat the mapping from any channel to subjective emotion as an open validity question.

We examine all three difficulties in a single archival dataset in which they can be examined at once. The data come from an experiment in which college students completed a 25-item untimed reasoning task while a commercial facial-analysis system logged its output in real time (Wang et al., 2021, 2025); data collection began in person and, interrupted by a university’s COVID-19 closure, resumed over Zoom half a year later, a design fact whose measurement consequences we document rather than set aside. The archive aligns person-level surveys (including math anxiety), item-level

behavioral logs, and a dense time-stamped stream of nine facial-inference channels recorded during the same items. The original publications used two channels (the joy- and anxiety-labeled outputs) as substantive emotion variables; here we consider the measurement questions that logically precede any such use of all nine.

Four questions organize the analysis, and we state them here in plain terms.

- (1) If a channel is scored item by item, do the resulting person differences replicate across the 25 items once the strong serial dependence of the stream is taken into account, including under a deliberately harsh accounting that credits each person-by-item cell with a single effective observation?
- (2) Does the answer depend on which channel is scored, and on which aspect of the stream (whether the channel is active at all, as against how large its output is when active)?
- (3) How much of what the nine channels record is shared, and can the shared part be attributed to anything more interpretable than the recording process itself?
- (4) Do the channel-based scores add information: do they improve held-out prediction of behavioral and survey outcomes beyond baselines built from the assessment’s own data?

The fourth question is framed as an interpretive boundary rather than as a primary claim: it asks whether the specific, coarse representations we evaluated add criterion information, not what facial process data could contribute in general.

The contribution is twofold. Substantively, we show that the answers dissociate: almost all of the channels order persons consistently across items, and this ordering survives severe discounting of the within-cell information, yet none of the evaluated aggregate representations improves any of the held-out benchmarks. Repeatable automated channels are not necessarily useful ones, and researchers who intend to use such channels require evidence on both questions. Methodologically, we assemble a reusable workflow for dense process channels, consisting of explicit measurement objects and provenance records, effective-information accounting under serial dependence, mixed-model generalizability with model-free corroboration, and held-out criterion benchmarks with matched folds. As far as we are aware, automated facial-inference channels have not previously been taken through this full chain within a single assessment dataset.

The plan of the article is as follows. In [Section 2](#) we describe the source study, the two administration modes, the archival reconciliation that defines the analytic

cohort, the nine channels and their measurement components, and the anatomy of the aligned stream. In Section 3 we map the four questions onto analyses: the effective-information scenarios, the generalizability models, the model-free split-half corroboration, the cross-channel structure analyses, and the two criterion benchmarks. Section 4 reports the results in the same order. In Section 5 we discuss the dissociation between repeatability and utility, reporting recommendations, and the limitations of the design, and Section 6 concludes.

2. The archival study

2.1. Source study and administration modes. The data come from an experiment on autonomy support and academic persistence conducted at a public university in the United States (Wang et al., 2021). In the published design, college students were randomly assigned to an autonomy-support manipulation (choosing between a mathematics and a literacy version of the task, as against being assigned one) and then completed 25 multiple-choice reasoning questions drawn from a Cultural Fair Intelligence Test item pool. The task was untimed. After an incorrect first attempt, the participant chose among three options: continue working on the item, request a hint, or skip the item; these behaviors, together with accuracy and time on task, formed the behavioral persistence measures of the original studies. A pre-task survey collected demographics and individual-difference scales, including the nine-item Abbreviated Math Anxiety Scale (Wang et al., 2025). A follow-up study of the mathematics-task subset related math anxiety to post-error skipping and help seeking (Wang et al., 2025). As described in the source publications, the 177 surveyed participants were recruited through the university’s research participation system; most were between 18 and 22 years of age (79.6%), the sample was predominantly female (89.8%), and a majority identified as White (81.9%) (Wang et al., 2021). The analytic cohort below is a subset of this sample, and the person-ordering claims of this article generalize, at most, over populations resembling it.

The administration was not conducted in a single mode, and this design fact matters for everything that follows. The source publications describe the procedure as an online study administered over Zoom during the university’s COVID-19 closure. The archival records preserve a more layered history: an in-person wave, in which participants were tested on site between February 18 and March 13, 2020, ended when the campus closed, and data collection resumed remotely over Zoom on September 2, 2020, continuing through January 29, 2021 (Figure 1a). The participant-level survey

records the mode explicitly (47 of the 177 surveyed participants are marked as tested on site), the two waves do not overlap in time, and no participant appears in both. We therefore describe the administration as the archive records it: a mixed-mode design with an in-person minority and an online majority, in which mode is completely confounded with calendar time. Throughout the article, mode is treated as a session-level description and as a rival explanation to be carried along, not as a manipulated factor.

Throughout the task, the Contain Intelligent Facial Expression Recognition System (CIFERS), a commercial system based on the Facial Action Coding System tradition, processed the participant’s face and emitted time-stamped numerical output (“ticks”) on multiple emotion-labeled fields. For the online sessions, the software read the participant’s face from the Zoom window on the study computer (Wang et al., 2021), so the online stream was computed from a compression-mediated video window fed by 127 participants’ own webcams of heterogeneous quality. The archival record does not document the capture path of the in-person sessions; direct camera capture is the natural configuration for the software, but it cannot be verified from the stored files, and we treat the two capture paths as different and only partly documented. The original publications describe the system as dividing the face into 47 units with a vendor-claimed capability of one observation per 50 ms (Wang et al., 2021). Independent technical documentation of the system’s training data, calibration, and score semantics was not available to us; this absence constrains the claims the channels can support, and we return to it in the Discussion. In the archival export analyzed here, the empirical median interval between successive ticks was approximately 160 ms (about 6 Hz), and the realized per-cell tick rate was essentially identical in the two administration modes (median 5.79 vs. 5.76 Hz per cell); the matched cadence shows that the software’s output rate did not differ by mode, not that the quality of its video input did. We treat the 50-ms figure as a nominal device description rather than an observed sampling rate. The present study is a secondary analysis of the de-identified archival research data; ethics approval and consent procedures for the original collection are described in the source publications.

2.2. Three data layers and the analytic cohort. The archive assembles three separately stored sources: the participant-level survey table underlying the published analyses, the item-level behavioral test logs, and the facial-inference database. Because each source has its own cohort, we report the cohort accounting as a reconciliation rather than as a single attrition funnel. The published survey sample comprised

177 participants. The combined cleaned test logs contained 179 participants with 4,475 canonical person-by-item rows (25 item positions each). The facial-inference database, after a historical rule that removed all sessions from duplicated database identifiers, contained 176 participants. The intersection of test logs and cleaned facial data, which we call the *matched cohort*, contained 169 participants and is the analytic cohort for all measurement analyses: 42 tested in person and 127 online (120 confirmed by the survey’s mode variable and 7 without survey records, all of whose sessions fall in the online wave). Criterion analyses use the participants with complete outcome and covariate data: $n = 169$ for the two log-derived outcomes and $n = 100$ for the three survey-dependent outcomes (Section 3.6). Full reconciliation tables, including the audited set differences between sources, are given in Appendix A.

Two further archival variables enter the analyses. The survey records the task version (mathematics or literacy; the 25-item positional structure is identical across versions), which serves as a covariate in the criterion benchmarks. The test logs also carry a historical internal label that reflects the data-collection path and file lineage rather than any substantive design factor; we preserve it as an archival covariate, document it in Appendix A, and do not interpret it. The archival cleaning pipeline, comprising session-level removal of duplicated identifiers, correction of the facial-database clock against the test-log clock, back-calculation of item start times, and assignment of ticks to item intervals, was reconstructed and audited before any analysis; its decisions originate in the original research team’s documented choices, and we treat them as fixed provenance that the present analyses neither relitigate nor silently modify (Appendix A).

2.3. Nine channels, their names, and their measurement components. The stream contains nine channels, listed in Table 1 with their basic descriptive properties. Six carry the emotion labels of the software’s output categories (Joy, Sad, Anger, Fear, Disgust, Contempt). The remaining three (Anxious, Attentive, Negative) carry names assigned in the original project’s analysis meetings, and the Negative channel is additionally a sign-folded absolute-magnitude transform, so its direction is not recoverable. We emphasize one point once, here, rather than repeating it at every mention: none of the nine names has been validated against subjective emotion in these data, by the vendor, by self-report, or by any external criterion. Channel names below are variable labels with a documented provenance, not construct claims; the complete provenance crosswalk, including the historical renaming and transform rules and the required first-mention wording, is recorded in Appendix B.

TABLE 1. The Nine Facial-Inference Channels and Their Sparsity

Channel	Active share of eligible ticks (%)	Cells with any variation (%)	Median lag-1 autocorrelation
Anxious ^a	38.1	91.4	.520
Attentive ^a	49.6	94.6	.512
Negative ^a	16.2	84.2	.532
Joy	0.4	12.0	.329
Sad	3.2	46.1	.373
Anger	2.5	46.5	.329
Fear	2.9	39.9	.206
Disgust	10.6	80.9	.352
Contempt	5.2	56.8	.495

Note. $n = 169$ participants; 4,126 observed person-by-item cells. “Active share” is the mean over cells of the proportion of eligible ticks at which the channel’s source value was nonzero; “cells with any variation” is the share of cells whose activation series is not constant; the median lag-1 autocorrelation is taken over cells with nonconstant series. ^aChannel name assigned in the original project’s analysis records rather than taken from the software’s output labels; the Negative channel is additionally a sign-folded absolute magnitude, so its direction is not recoverable. No channel name is validated here as a measure of the corresponding subjective emotion; the full provenance crosswalk and naming history are given in Appendix B.

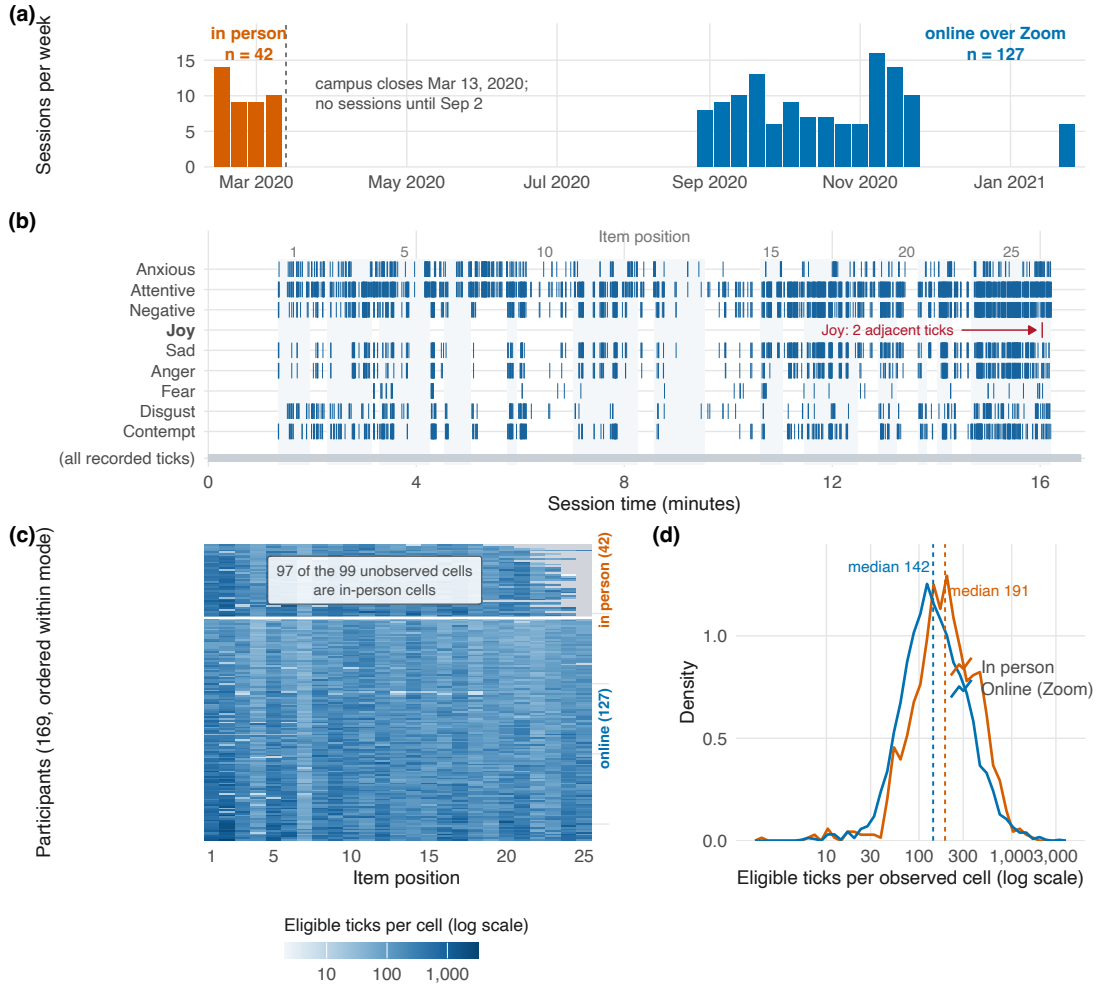
For each channel we distinguish three components of the tick stream within a person-by-item cell (the unit formed by one participant working on one item). *Activation*, the primary measurement object, is the binary event that the software emitted a nonzero source value at an eligible (recorded and item-aligned) tick; the cell-level activation score is the proportion of eligible ticks that were active. Activation records the production of signal by the software and carries no implication that an emotion occurred. Two secondary components decompose what happens when a channel is active: the *active boundary* (among active ticks, whether the transformed magnitude exceeds its floor) and the *strict-positive magnitude* (the transformed magnitude when it does). This hurdle-style decomposition separates how often a channel is active from how large it is when active, two properties with different sparsity and, as shown below, different repeatability. Unrecorded ticks are treated as missing observation opportunities, never as zeros. Table 1 already displays the fact that drives much of what follows: the channels differ enormously in sparsity, with the Joy channel active on only 0.4% of eligible ticks and varying within only 12.0% of cells.

2.4. Anatomy of the aligned stream. Figure 1 shows what the assembled dataset looks like. Of 964,143 recorded ticks for the matched cohort, 896,309 fall within an

item interval and enter the person-by-item accounting; ticks outside item intervals are retained only in session-level recording covariates. The specimen panel (Figure 1b) displays a single participant’s session and conveys the texture of the data more directly than any summary: activation arrives in bursts, long stretches of a channel are entirely silent, the Attentive and Anxious channels are active for much of the session, and the Joy channel is active at only two adjacent ticks of the 5,119, a single brief event late in the session. The session shown is not an unusual one; it was selected deterministically as the participant whose total aligned tick count equals the cohort median (5,119 ticks) with all 25 cells observed.

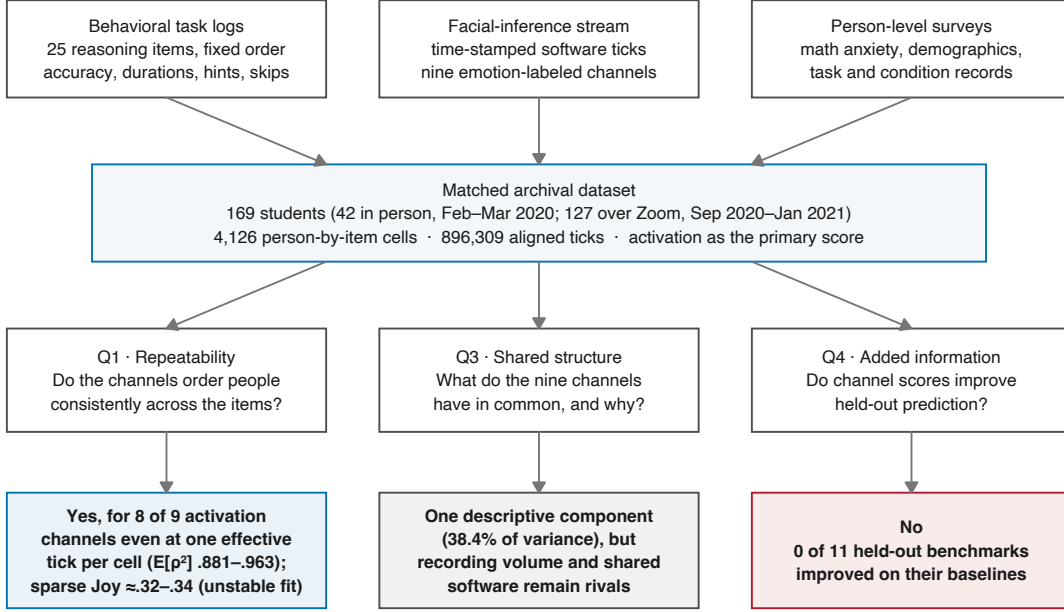
For the 169 participants, $169 \times 25 = 4,225$ person-by-item cells are possible and 4,126 (97.7%) contain an aligned facial-inference summary. Observed cells are dense: a mean of 217.2 and a median of 153 eligible ticks per cell. The 99 unobserved cells are not scattered at random, and their pattern is a mode fact (Figure 1c): 97 of the 99 belong to in-person participants, and the cells concentrate at the late item positions (of the 99, 42 fall at item 25, 26 at item 24, and 15 at item 23), a pattern consistent with facial recordings that ended before the participant finished the last items. Forty-one of the 42 in-person participants have at least one unobserved cell, whereas online coverage is nearly complete (3,173 of 3,175 possible cells). The in-person sessions are also somewhat longer at the item level (median 34 vs. 25 seconds per item), so their observed cells carry more ticks (median 191 vs. 142), and the in-person wave contributes 241,119 of the 896,309 aligned ticks. Missing cells are never imputed and never entered as zeros; analyses use the realized coverage pattern, and the mode-linked coverage asymmetry is one of the session-level facts carried into the interpretation of every result below.

FIGURE 1. Anatomy of the Archival Stream



Note. (a) Sessions per week for the 169-participant matched cohort. The in-person wave (orange) ends at the COVID-19 campus closure on March 13, 2020; the online wave (blue) runs from September 2, 2020 to January 29, 2021. (b) One participant's session (online wave; selected deterministically as the participant with all 25 cells observed whose total aligned tick count equals the cohort median of 5,119). Each vertical mark is one tick at which the named channel was active; alternating background bands mark the 25 item intervals; the bottom strip shows every recorded tick, including those between items. The Joy channel (red) is active at only two adjacent ticks, which render as the single marked event. (c) Eligible ticks per person-by-item cell for all 169 participants (rows, grouped by administration mode and ordered by total aligned ticks within mode) across the 25 item positions; grey cells are the 99 unobserved cells. (d) Distribution of eligible ticks per observed cell by mode (medians 191 in person, 142 online; both about 5.8 Hz).

FIGURE 2. From Archive to Questions to Answers



Note. Three archival layers are reconciled into the matched dataset (Section 2.2), on which three families of analyses answer the questions of the Introduction (the component question, Q2, reuses the Q1 machinery). The findings shown in the bottom row are established in Section 4.

3. From questions to analyses

3.1. Overview. Figure 2 summarizes the architecture of the analysis, and Table 2 states, for each of the four questions of the Introduction, the measurement object it is asked of, the analysis that answers it, and where the answer is reported. Two design commitments run through everything. First, no analysis credits the 896,309 aligned ticks with independent information; every repeatability statement is made under an explicit accounting of how much information the autocorrelated stream actually contains. Second, repeatability and utility are evaluated by different logics on the same data, so that neither answer is permitted to stand in for the other. One scope note belongs here as well: the randomized autonomy-support manipulation of the source studies enters the present analyses only as a covariate; channel sensitivity to that manipulation is a substantive question owned by the source studies, not a measurement property evaluated in this article.

3.2. Serial dependence and effective information. Within a person-by-item cell, adjacent ticks are strongly autocorrelated: across channels, the median lag-1 autocorrelation of the activation series ranges from .206 to .532 (Table 1). Autocorrelation

TABLE 2. How Each Question Is Answered

Question	Measurement object	Analysis	Reported output
Q1. Do person differences replicate across items?	Cell-level activation counts (per channel)	Binomial mixed model per channel; generalizability coefficient under three effective-information scenarios (Section 3.3)	$E[\rho^2]$ per channel and scenario (Table 3, Figure 3)
	Person half-scores over random item halves	500 unequal (12/13) item splits; Spearman correlation with Spearman–Brown adjustment (Section 3.4)	Median adjusted stability and split-to-split range (Table 4, Figure 4)
Q2. Does the answer depend on the component scored?	Active-boundary and strict-positive-magnitude cell scores	Same mixed-model and split-half machinery applied per component, with a sparse-support reporting policy	Component-by-channel stability matrix (Figure 4b; Appendix D)
Q3. What do the channels share, and why?	Nine person-level activation scores	Score correlations; descriptive principal components; associations with session recording covariates and administration mode (Section 3.5)	Correlation structure, PC1 share and loadings, recording-volume and mode associations (Figure 5)
Q4. Do the scores add held-out information?	Four aggregate person representations; nine concurrent whole-item features	Repeated cross-validation against prespecified baselines, folds drawn at the participant level (Sections 3.6 and 3.7)	Held-out ΔR^2 (five person-level outcomes) and relative loss reduction (six item-level outcomes) (Tables 5 and 6, Figure 6)

Note. Q1–Q4 are the four questions stated at the end of the Introduction. All analyses operate on the 169-participant matched cohort of Section 2.2; the criterion benchmarks use the complete-data subsets given in Section 3.6.

of this size changes what a tick count means. For a lag-1 autoregressive process, the variance of a cell mean based on m ticks equals the variance of a mean of

$$m_{\text{eff}} = m \frac{1 - \hat{\rho}_1}{1 + \hat{\rho}_1} \quad (1)$$

independent ticks, so a cell with the average of 217.2 eligible ticks carries the information of at most roughly 66 to 143 independent observations depending on the channel, between 31% and 66% of the nominal count (derivation and the general dependence formula in Appendix C). Because even this correction assumes a specific dependence structure, and residual dependence beyond lag 1 would reduce the counts further, our headline scenario is more conservative still: it credits each cell with a *single* effective tick and asks whether person ordering survives when within-cell information is discounted entirely. We stress the roles of the three scenarios. The one-tick scenario is a conservative stress test rather than a formal lower bound; the AR(1) scenario of Equation (1) is a secondary approximation; and the nominal-count scenario, which assumes independent ticks, serves only as a reference against which the cost of the independence assumption can be seen.

3.3. Generalizability models and coefficients. For each channel, we fit a binomial generalized linear mixed model to the cell-level activation counts of the 169-person cohort (Bates et al., 2015): active ticks out of m_{pi} eligible ticks in the cell of person p and item position i , with a logit link, fixed effects for the 25 item positions, and random intercepts for persons and person-by-item cells,

$$\text{logit}(\pi_{pi}) = \beta_i + u_p + w_{pi}, \quad u_p \sim N(0, \sigma_p^2), \quad w_{pi} \sim N(0, \sigma_{pi}^2). \quad (2)$$

On the latent (logit) scale this decomposes variance into a person component σ_p^2 , a person-by-item component σ_{pi}^2 , and the standard logistic tick-level residual $\pi^2/3$ (Nakagawa & Schielzeth, 2010; Nakagawa et al., 2017). At the estimation stage the binomial likelihood treats ticks as conditionally independent trials given the cell effect, with within-cell dependence absorbed by the person-by-item intercept; the information scenarios below then discount the tick-level contribution rather than re-model the serial process. The generalizability coefficient for a relative (person-ordering) decision over the 25-item-position design is then

$$E[\rho^2] = \frac{\sigma_p^2}{\sigma_p^2 + \frac{\sigma_{pi}^2}{25} + \frac{\pi^2/3}{25 m_{\text{eff}}}}, \quad (3)$$

with m_{eff} set to 1 (stress test, headline), to the value of Equation (1) (AR(1), secondary), or to m (nominal reference). The coefficient has a plain reading: it asks how consistently the average of a person’s 25 cell scores would order that person against the others, charging person-by-item idiosyncrasy and tick-level noise as error, with

the tick-level term discounted according to the information scenario. Secondary components are modeled analogously (binomial models for the active boundary among active ticks; a Gaussian model for strict-positive magnitudes on the historical log scale).

Four scope conditions attach to every coefficient. First, the coefficients are relative, person-ordering coefficients; we do not report absolute-decision dependability, which would require absolute-level interpretations that these channels do not have. Second, they are defined on the component-model latent scale and are reported as point estimates without interval statements. Third, item content and presentation position are completely confounded by the fixed administration order, so the item facet is always item/position, and treating the 25 observed positions as exchangeable draws from a larger universe is a model-based idealization; the decision the coefficient is meant to inform is therefore the generalization of a person’s average score over administrations of forms like this one, while the split-half design of the next subsection carries the concrete same-form referent. Fourth, all coefficients are conditional on the observed single-session design and its realized coverage: the models pool the two administration modes, any mode-linked differences in level are absorbed into the person component, and nothing here estimates cross-session stability or separates stable person characteristics from stable session conditions. Model estimation details, variance-component tables, and one estimation note on the sparse Joy fit are given in Appendix C.

3.4. Model-free corroboration: repeated item split-halves. Because Equation (3) depends on a parametric variance decomposition, we corroborated it with a design that involves no model. In each of 500 replications, the 25 item positions were randomly partitioned into halves of 12 and 13; each participant’s half-scores were the means of their observed cell-level component values within each half (requiring at least three observed cells per half); and the replication statistic was the Spearman rank correlation between half-scores across participants (computed when at least 20 participants had valid scores in both halves). We report the median raw half-correlation and the Spearman–Brown adjusted value $2r/(1+r)$, a heuristic on two counts (the halves are unequal, and the classical correction is derived for Pearson correlations of parallel composites rather than for rank correlations), together with the 2.5th–97.5th percentile range across replications; that range is an empirical stability range across item splits, not a confidence interval. Under a support policy fixed

after external review and before drafting, adjusted values are suppressed from headline reporting for any channel-component whose median supported sample falls below 40 participants; raw statistics and support counts are always reported (Appendix D).

3.5. Cross-channel structure. Person-level activation scores for the nine channels are the estimated person intercepts (conditional modes) from the activation models; a raw-score counterpart closely reproduces their ordering (Appendix E). We summarize cross-channel structure descriptively, using the score correlation matrix and a principal component analysis of the standardized scores (Jolliffe & Cadima, 2016), with the first component (PC1) sign-aligned so that its loading sum is positive. PC1 is used as a descriptive summary of cross-channel commonality and, later, as the “general activation score” input to the criterion benchmark. It is not interpreted as a general emotion factor: all nine channels share one software pipeline, one camera path, and one session per person, so common algorithmic processing, individual expressivity, camera and lighting conditions, and recording volume are all rival explanations for shared variance (Podsakoff et al., 2003). We probe the recording-volume explanation directly by correlating PC1 with three session-level recording covariates (the proportion of possible cells observed, the log of total recorded ticks, and the proportion of ticks not assignable to any item interval), and we report the administration-mode composition of the same score as an archival descriptive, since mode is a session-level condition bundled into the same rival-explanation family.

3.6. Exploratory held-out person-level criterion benchmark. The criterion analyses address a narrow question: whether coarse aggregate person representations of the facial channels improve out-of-sample prediction of specified outcomes beyond specified baseline covariates. They are exploratory benchmarks, not preregistered confirmatory tests, and their baselines were fixed in advance by the historical survey analysis tables.

Five person-level outcomes were evaluated. (a) *Math anxiety* (Abbreviated Math Anxiety Scale mean; $n = 100$), with baseline covariates task accuracy, mean response time, sex, a three-category archival mathematics self-report item (“do you consider yourself good at math”: no / yes / it depends), and experimental condition. (b–c) *Post-error help rate* and *post-error skip rate* (proportions of post-error opportunities; $n = 100$), with the same baseline plus math anxiety. (d–e) *First-attempt accuracy rate* and *mean log item duration* ($n = 169$), with baseline covariates the task-version indicator and two recording covariates (coverage proportion and log total ticks), so

that for these two outcomes the baseline already absorbs recording volume. Because ticks accrue at an approximately constant rate, log total ticks is close to a proxy of total recorded time, so the duration benchmark is volume-controlled by construction and is a deliberately hard target for any additional predictor. Following external review, the three-category self-report item enters all models as a nominal factor with both dummy columns retained in every fold, a correction of a historical path that had treated the categories as a continuous score; the correction changed baseline performance but not the direction of any facial increment (Appendix F).

Four facial representations were added to each baseline: (a) the *general* activation score (PC1); (b) general plus an *Anxious-specific* score (the residual of the Anxious activation score regressed on PC1), motivated by the anxiety-adjacent role of that channel in the source studies; (c) *all nine* activation scores jointly, an intentionally demanding sensitivity model given nine correlated predictors at $n = 100$; and (d) the general score *residualized on the three recording covariates*, with the residualization learned on training folds only, as a nuisance-adjusted sensitivity. One boundary of this design should be stated explicitly: the PC1 loadings and the Anxious-specific residualization are outcome-blind transforms estimated once on all 169 participants before cross-validation, not refitted inside each training fold. No criterion outcome enters these transforms, so they cannot leak outcome information, but the reported cross-validation evaluates prediction conditional on this full-cohort representation rather than an end-to-end pipeline deployed to unseen persons; we expect this shared representation, if anything, to give the facial models a stabler and more favorable footing than fold-refitted transforms would.

Each outcome used 100 repetitions of five-fold cross-validation with folds drawn at the participant level and shared by all models within a repetition, so that representation contrasts are within-fold. Predictors were standardized using training-fold statistics only. Performance is held-out R^2 , and we report the median and 2.5th–97.5th percentile range of baseline R^2 and of each representation’s increment ΔR^2 across the 100 repetitions. These ranges describe stability across repeated data splits; they are not confidence intervals, and a negative median ΔR^2 is evidence of no detectable improvement for the evaluated representation under these learners and samples, not evidence of equivalence or of harm.

The task covariate deserves one further remark, since an external pre-submission audit found and corrected a contract violation in it. The task variable in every criterion baseline is the validated survey task assignment (mathematics vs. literacy); the

historical data-collection file label, which disagrees with the survey assignment for 35 of the 162 participants with known task, is never substituted for it. All eleven benchmarks reported here were rerun under the contracted variable. Seven participants lack a survey task value; under the frozen primary policy they are retained with an explicit missing-task indicator, and a complete-case sensitivity that drops them instead reproduces every conclusion (Appendix F). The three survey-outcome benchmarks are estimated, as fixed in the frozen analysis plan, on the mathematics-task subsample selected by the validated survey assignment.

3.7. Concurrent whole-item criterion benchmark. A parallel benchmark asks the utility question at a finer grain: whether the nine cell-level activation proportions for an item, which are concurrent whole-item summaries of the facial stream during that item, improve prediction of six item-level outcomes (first-attempt correctness, hint request, skip, log item duration, and post-error help and skip; 4,126 cells, 1,722 post-error cells). Baseline models contain item-position fixed effects and the task-version indicator; the facial model adds the nine features standardized on training folds. Folds are drawn at the participant level (five-fold, 50 repetitions), so that prediction is always for unseen persons. Loss is the mean negative Bernoulli log-likelihood for binary outcomes and root mean squared error for duration, and we report the median relative loss reduction (baseline minus facial, as a proportion of baseline loss) with its repeated-split range. The benchmark evaluates concurrent whole-item aggregates only; it does not test event-locked or within-item dynamic representations, and no conclusion below extends to them.

3.8. Provenance, external review, and quality assurance. Reproducibility operates at three distinct levels, which we state separately. The released numerical artifacts are internally consistent and re-derivable from the frozen releases; the consumer and quality-assurance environment that reads those releases is documented by the package dependency contract and a session snapshot, without a demonstrated exact clean restore; and the R and package versions used for the original model fits were not recorded at fit time and are documented as unrecorded. The full pipeline was subjected to two external code-and-claims reviews before submission; the first (30 findings) prompted the coding correction, the sensitivity representations, the effective-information reporting hierarchy, and the sparse-support policy described above, and the second found the task-covariate contract violation whose correction is described in Section 3.6. A statistical-content quality-assurance release re-derives the registered

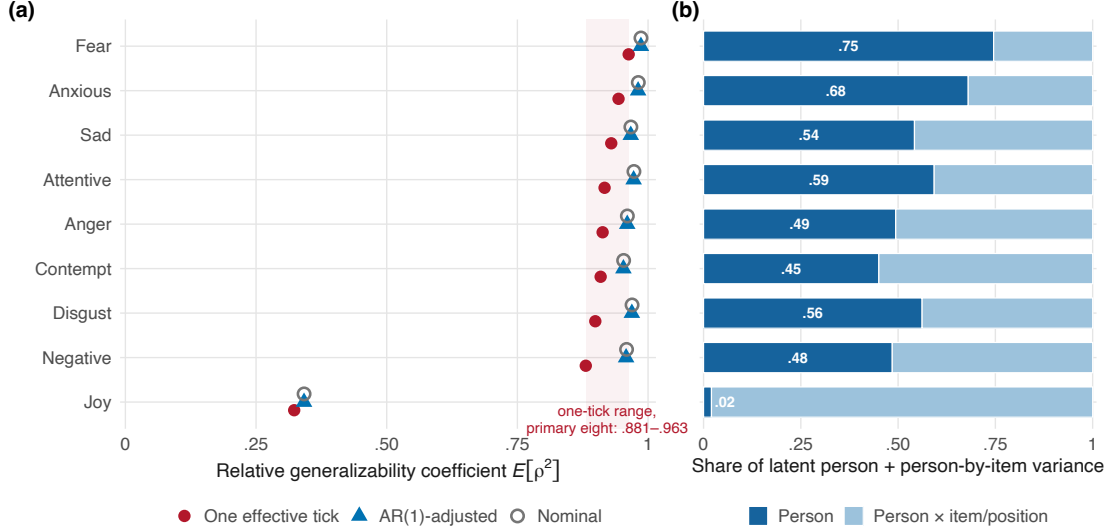
manuscript quantities from the frozen artifacts and verifies 18 machine checks (summarized in Appendix G). The public package separately checks 34 headline registry keys and classifies all 35 contracted manuscript targets; these are declared verification scopes rather than claims to cover every printed token. The manuscript’s six core claims are each bound to specific release files in a machine-readable claim–evidence matrix. Appendix G documents the reviews, the checks, the claim matrix, and the release hashes. The administration-mode facts of Section 2, which postdate the frozen releases, are archival descriptives re-derived at build time from the immutable stores by a versioned script and checked by the same number-verification gate (Appendix G).

4. Results

4.1. Repeatability under information discounting. The first question is whether person differences in each channel replicate across the 25 item positions once the serial dependence documented in Section 2.4 is taken into account. Table 3 reports the relative generalizability coefficients under the three effective-information scenarios, and Figure 3a displays them. Under the nominal independence reference, all channels except Joy show $E[\rho^2] \geq .953$; however, that scenario counts every tick as new information and is reported only as a reference. Of principal interest is the one-effective-tick stress test. When each of a person’s 25 cells is credited with a single effective observation, the eight channels other than Joy retain $E[\rho^2]$ between .881 (Negative) and .963 (Fear). Person ordering in these activation channels therefore does not depend on treating dense ticks as independent; it is preserved even when within-cell information is discounted entirely. The scenarios rescale the residual averaging benefit while holding the fitted variance components fixed; they are decision-study sensitivities, not refits under an alternative serial process.

Joy, in contrast, is low under every scenario (.323 under the stress test and .342 at nominal density), so its problem is not the discounting of within-cell information. One estimation caveat attaches to its exact values: at Joy’s extreme sparsity the fit is optimizer-sensitive, and Appendix C cautions against interpreting the model-based coefficient precisely; the model-free split-half evidence of the next paragraph, which puts Joy at .703, is the corroborating anchor for the qualitative conclusion that Joy is the unstable channel. Figure 3b shows the mechanism: for the primary eight channels the person component accounts for .45 to .75 of the latent person-plus-interaction variance, whereas for Joy it accounts for only .02, the pattern expected of a channel whose rare activations are idiosyncratic to particular person-item combinations rather

FIGURE 3. Activation Generalizability Under Effective-Information Scenarios



Note. (a) Relative $E[\rho^2]$ per channel (ordered by the stress-test value) under the nominal independence reference, the AR(1) effective-tick approximation, and the one-effective-tick-per-cell stress test; the shaded band marks the stress-test range of the primary eight channels (.881–.963).

(b) Share of the latent person-plus-interaction variance attributable to the person component: Joy’s low coefficients reflect interaction-dominated variance (.02 person share), not within-cell information loss.

than stable person differences. Joy is also the channel that Table 1 already marked as extreme: its activation series varies in only 12.0% of cells. Its behavior is thus informative rather than anomalous, and we return to it below.

The mixed-model coefficients could in principle reflect parametric assumptions, so the second half of the answer comes from the design that involves no model. Across 500 random 12/13 splits of the item positions (Table 4; Figure 4a shows the full distributions), median raw half-correlations for activation range from .542 (Joy) to .964 (Anxious), with Spearman–Brown-adjusted medians of .703 to .982 and finite statistics in all 500 splits for every channel. Because half-scores average cells at their realized tick density, the matched model-based comparison is the nominal column of Table 3, relative to which the rank-based adjusted values run somewhat lower (by roughly .05 to .09 for Sad, Anger, Fear, and Contempt), as expected when an observed-scale rank statistic is set against a latent-scale coefficient. The two designs agree on the conclusion that matters: eight channels order persons highly consistently across item splits, and Joy does not. That a parametric latent-scale coefficient and a

TABLE 3. Relative Generalizability Coefficients $E[\rho^2]$ for Activation Under Three Effective-Information Scenarios

Channel	Median lag-1	AR(1) m_{eff}	$E[\rho^2]$ by scenario		
			One tick	AR(1)	Nominal
Anxious	.520	68.6	.943	.981	.981
Attentive	.512	70.2	.917	.972	.973
Negative	.532	66.4	.881	.958	.959
Joy	.329	109.7	.323	.342	.342
Sad	.373	99.3	.930	.967	.967
Anger	.329	109.6	.913	.960	.960
Fear	.206	143.1	.963	.986	.986
Disgust	.352	104.2	.899	.969	.969
Contempt	.495	73.4	.909	.953	.953

Note. $n = 169$ participants; 4,126 person-by-item cells; mean 217.2 eligible ticks per observed cell. Coefficients are relative (person-ordering) generalizability coefficients on the component-model latent scale, from binomial mixed models with 25 item-position fixed effects and random person and person-by-item intercepts (Equation 3); they are point estimates without interval statements. “One tick” credits each cell with one effective observation (headline stress test; not a formal lower bound); “AR(1)” uses Equation 1; “Nominal” assumes independent ticks (reference only). All values are conditional on the observed 25-item-position task and realized coverage; extrapolation to exchangeable item/positions is model-based, and no cross-session claim is made. Variance components are given in Appendix C.

model-free rank statistic converge on this ordering conclusion provides the strongest evidence in these data that the person-level repeatability of the eight activation channels is not an artifact of either method.

4.2. Heterogeneity across measurement components. The second question is whether repeatability attaches to a channel as such or to a particular scoring of it. The secondary components answer clearly: reliability evidence obtained for one component of a channel does not transfer to another (Figure 4b; full table in Appendix D). For some channels, occurrence and conditional magnitude are both stable (Anxious: boundary adjusted median .956, magnitude .951); for others, activation is stable while the boundary component falls sharply (Anger: activation .867 vs. boundary .237). Joy illustrates the interaction of sparsity with estimation. Its boundary and magnitude components have median supported samples of only 24 and 20 participants (with finite statistics in 494 and 285 of 500 splits), below the 40-person support threshold, so their adjusted values are suppressed from headline reporting under the prespecified policy. It should be noted what this result does not assert:

TABLE 4. Repeated Unequal (12/13) Item Split-Half Stability of Activation Person Scores

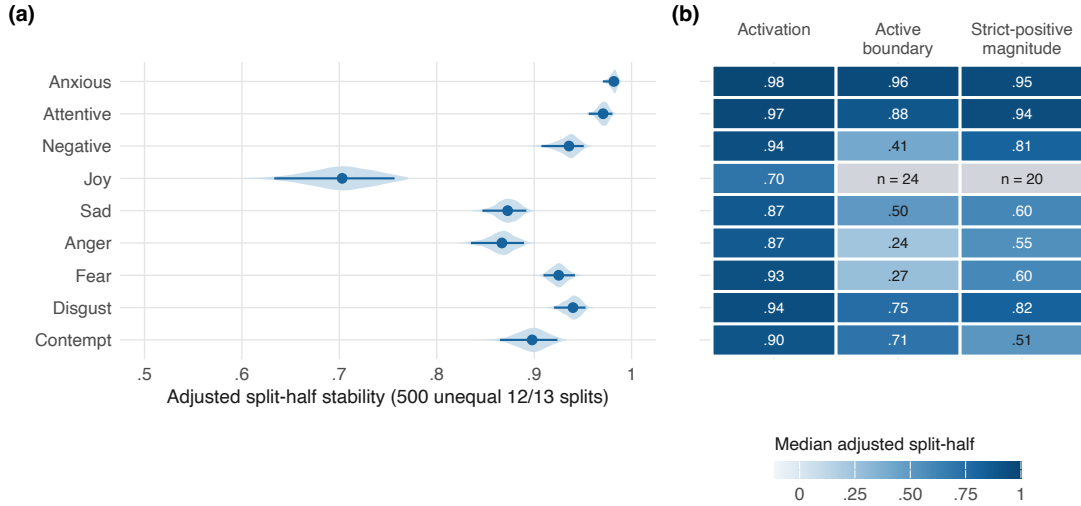
Channel	Raw half ρ	Adjusted	2.5%–97.5% range	Finite/attempted
Anxious	.964	.982	[.971, .987]	500/500
Attentive	.943	.971	[.956, .980]	500/500
Negative	.879	.936	[.907, .951]	500/500
Joy	.542	.703	[.633, .757]	500/500
Sad	.774	.873	[.847, .892]	500/500
Anger	.765	.867	[.835, .890]	500/500
Fear	.861	.925	[.910, .942]	500/500
Disgust	.887	.940	[.920, .953]	500/500
Contempt	.815	.898	[.865, .924]	500/500

Note. Medians across 500 replications; median supported $n = 169$ in every replication. “Raw half ρ ” is the Spearman correlation between person half-scores; “Adjusted” applies the Spearman–Brown formula $2r/(1+r)$, a heuristic approximation for unequal 12/13 halves; the bracketed range is the 2.5th–97.5th percentile of the adjusted statistic across replications (an empirical stability range across item splits, not a confidence interval). Secondary components (active boundary, strict-positive magnitude) are reported in Appendix D.

Joy’s activation component is estimable and simply lower (.703 adjusted; .323 under the one-tick stress test), and nothing here shows the channel to be malfunctioning. Joy is an empirically sparse, sparsity-sensitive counterexample, and it demonstrates that channel-level heterogeneity is too large for the nine channels to be averaged into a single undifferentiated “facial signal.”

4.3. Cross-channel structure and its rival explanations. The third question concerns what the nine channels share. The nine activation person scores are far from independent (Figure 5a). The correlation structure shows an Anger–Sad–Negative–Contempt cluster ($r = .50$ –.74), a very high correlation within the project-named pair Anxious–Attentive ($r = .92$), and a Joy score modestly related to the first cluster and essentially unrelated to the second. The first principal component of the standardized scores summarizes 38.4% of total variance (PC2: 24.8%; PC3: 11.9%; Figure 5b), with dominant loadings from Anger (.478), Negative (.473), Sad (.428), and Contempt (.416) and near-zero loadings for Anxious (−.029) and Attentive (.087; Figure 5c). Descriptively, PC1 is an Anger/Sad/Contempt-and-Negative-dominant activation signature, and the sharp separation between that cluster and the Anxious–Attentive pair is consistent with distinct processing families inside one software pipeline (the channels’ provenance history, documented in Appendix B, points the same way).

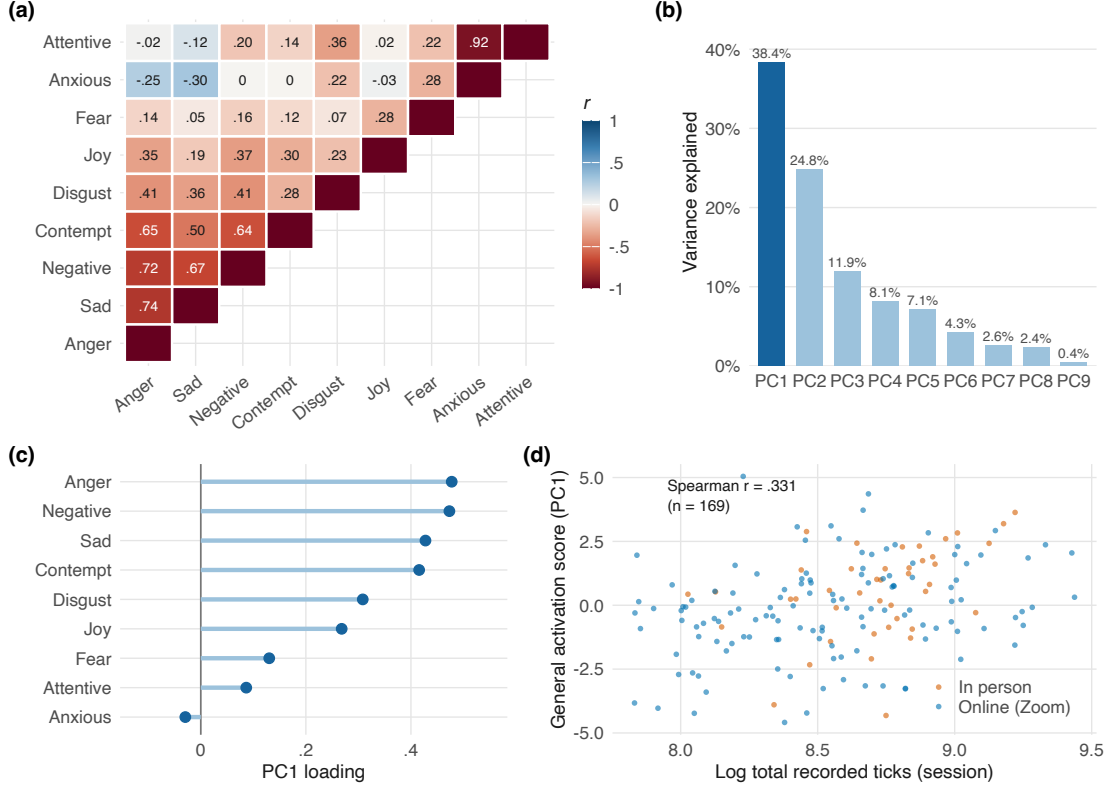
FIGURE 4. Model-Free Repeated Item Split-Halves



Note. (a) Full distributions of the Spearman–Brown-adjusted statistic across the 500 unequal 12/13 item splits for activation (violins), with the median and the 2.5th–97.5th percentile stability range overlaid; Joy is the low, wide outlier. (b) Median adjusted statistic for all nine channels and all three measurement components; the two grey cells are Joy’s secondary components, whose adjusted values are suppressed under the sparse-support policy (median supported samples of 24 and 20 participants).

However, PC1 cannot be interpreted as a general emotion factor or a negative-affect factor, and the session-level covariates indicate why caution is warranted. PC1 correlates with log total recorded ticks (Spearman $r = .331$; Figure 5d), weakly and negatively with coverage proportion ($-.164$), and near zero with the unassigned-tick proportion ($.076$; $n = 169$; Appendix E). Part of the shared variance therefore reflects how much the session recorded. The administration modes of Section 2.1 sit visibly inside this association: in-person items ran longer and therefore accumulated more ticks, and the general activation score runs higher in the in-person group (mean 0.54, vs. -0.18 online, in score units; Appendix E). The mode composition is channel-specific rather than a single common shift: standardized differences in the activation person scores reach $d \approx 0.6$ – 0.7 for Anxious, Attentive, Fear, and Disgust and are near zero for Sad, Anger, and Contempt (Appendix E), so the association is not confined to the recording-volume component. Its consequence for the repeatability claims is bounded: treating each channel’s score-scale mode share as if it were latent person variance and removing it lowers the one-effective-tick coefficients of the primary eight channels by at most .007 (Appendix E). We report all of this as archival

FIGURE 5. Cross-Channel Structure of the Activation Person Scores and Its Session-Level Correlates



Note. (a) Correlations among the nine activation person scores (cluster-ordered): an Anger–Sad–Negative–Contempt cluster, a very high Anxious–Attentive correlation, and a largely separate Joy score. (b) Variance explained by the nine principal components. (c) PC1 loadings: the descriptive common component is Anger/Sad/Contempt-and-Negative-dominant, with near-zero Anxious and Attentive loadings. (d) The general activation score (PC1) against log total recorded ticks per session (Spearman $r = .331$, $n = 169$), with points colored by administration mode: in-person sessions concentrate at higher recording volume and higher PC1, one strand of the session-condition rival-explanation family.

description, not as effects of mode: mode is confounded with calendar time, camera path, and every other session-level condition. Because every channel also shares the same software, camera, and session, common algorithmic processing and stable individual expressivity remain unseparated rival explanations; with one session per person, the design cannot distinguish them. The commonality is established as a statistical description but remains unidentified as a construct.

4.4. The criterion benchmarks. The fourth question is whether any of this repeatable signal adds held-out information. Table 5 reports the person-level benchmark,

and Figure 6 displays it. Baseline performance varied widely by design (Figure 6a): the baseline explained a median held-out R^2 of .771 for mean log duration and .692 for post-error skip rate, but only .134 for math anxiety, $-.020$ for first-attempt rate, and $-.032$ for post-error help rate, so the benchmark spans outcomes with ample unexplained variance and outcomes for which the baseline left little scope. The two outcomes with negative baseline medians are correspondingly weak tests: when the baseline predicts worse than the outcome mean at this sample size, an increment is a contrast against a failing model, and a null there says little about the channels. Across all five outcomes and all four representations, no evaluated representation produced a stable positive held-out improvement: median ΔR^2 ranged from $-.006$ to $-.021$ for the general score, from $-.008$ to $-.032$ for the specific representation, and from $-.026$ to $-.137$ for the all-nine representation, and in no outcome-representation pair did the facial model improve held-out R^2 in even a third of the repetitions (maximum proportion .29). The all-nine model’s larger deficits should be read in the light of its design: nine correlated predictors at $n = 100$ make it a demanding sensitivity model, and its behavior is consistent with the predictor-count burden. We do not interpret negative increments as evidence that the facial features worsen behavior.

The nuisance-adjusted sensitivity bounds one alternative explanation. If the general score’s criterion behavior were driven by the recording-volume information documented in Section 4.3, removing training-fold-estimated recording variance should change the result; in fact the largest absolute change in any median ΔR^2 was .003. Recording-volume information is present in the general score, but adjusting for it neither improved nor reversed the criterion conclusion.

The item-level benchmark reaches the same conclusion at a finer grain (Table 6; Figure 6c). Adding the nine concurrent whole-item activation features to the item-position and task-version baselines reduced held-out predictive loss for none of the six outcomes: median relative loss reductions ranged from $-.004$ (first-attempt correctness) to $-.026$ (post-error skip), with the entire repeated-split range below zero for every outcome. Two boundaries attach to this result. It concerns concurrent whole-item aggregates (event-locked, within-item, and dynamic representations were not evaluated), and it does not support causal or mechanistic conclusions about the loss increase, which is consistent with adding nine noisy correlated predictors to saturated item-position baselines.

TABLE 5. Exploratory Held-Out Person-Level Criterion Benchmark:
Median Baseline R^2 and Median Increments ΔR^2

Outcome	n	Baseline R^2	General	Specific	All nine	Nuis.-adj.
First-attempt rate	169	-.020	-.007	-.009	-.062	-.007
Math anxiety	100	.134	-.011	-.016	-.112	-.011
Mean log duration	169	.771	-.006	-.008	-.026	-.003
Post-error help rate	100	-.032	-.021	-.032	-.137	-.018
Post-error skip rate	100	.692	-.007	-.010	-.040	-.007

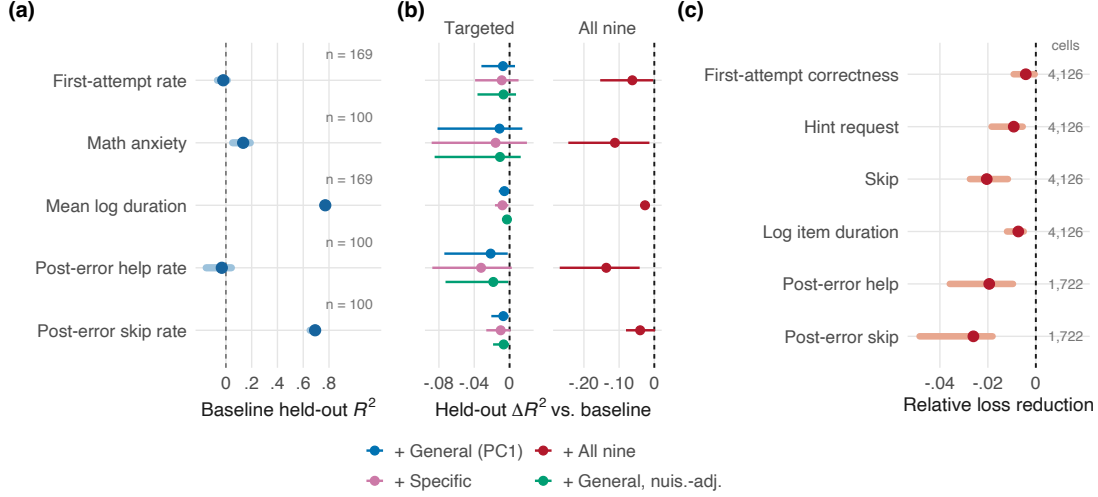
Note. Medians across 100 repetitions of participant-level five-fold cross-validation with folds shared by all models within a repetition; predictors standardized, and the nuisance residualization estimated, on training folds only. Baselines: math anxiety—accuracy, mean response time, sex, the three-category mathematics self-report factor, condition; post-error outcomes—the same plus math anxiety; first-attempt rate and mean log duration—the survey task version with its frozen missing-task indicator, coverage proportion, and log total ticks. Representations: General = activation PC1; Specific = PC1 + Anxious-specific residual; All nine = nine activation scores; Nuis.-adj. = PC1 residualized on three recording covariates. All values are from the task-corrected release; the survey-outcome rows ($n = 100$) are numerically unchanged by the correction, and no conclusion direction changed anywhere (maximum median shift .005; complete-case sensitivity in Appendix F). The benchmark is exploratory; repeated-split 2.5%–97.5% ranges for every cell are in Appendix F, and no median increment is positive. Negative increments indicate no improvement for the evaluated representations under these benchmarks; they are not evidence of equivalence, harm, or general nonutility.

TABLE 6. Concurrent Whole-Item Activation Benchmark: Held-Out
Relative Loss Reduction Versus Baseline

Outcome	Family	Cells	Median RLR	2.5%–97.5% range
First-attempt correctness	Binomial	4,126	-.004	[-.009, -.000]
Hint request	Binomial	4,126	-.009	[-.018, -.005]
Skip	Binomial	4,126	-.020	[-.027, -.012]
Log item duration	Gaussian	4,126	-.007	[-.012, -.005]
Post-error help	Binomial	1,722	-.019	[-.036, -.010]
Post-error skip	Binomial	1,722	-.026	[-.048, -.018]

Note. RLR = relative loss reduction, (baseline loss – facial-model loss)/baseline loss, so positive values would favor the facial model; medians and 2.5th–97.5th percentile ranges across 50 repetitions of participant-level five-fold cross-validation ($n = 169$ participants). Loss is mean negative Bernoulli log-likelihood (binomial outcomes) or root mean squared error (duration). Baselines contain item-position fixed effects and the survey task-version indicator with its frozen missing-task indicator (task-corrected release); the facial model adds the nine cell-level activation proportions standardized on training folds. The first-attempt upper range endpoint is -.0005, below zero at higher precision. In zero of six outcomes did the facial model improve held-out loss in more than half the repetitions. The benchmark evaluates concurrent whole-item aggregates only.

FIGURE 6. Held-Out Criterion Benchmarks



Note. (a) Baseline held-out R^2 per person-level outcome (points: medians across 100 repetitions; bands: 2.5th–97.5th percentile repeated-split ranges), showing the wide range of baseline performance the benchmark spans. (b) Held-out ΔR^2 relative to baseline for the four facial representations, drawn in two sub-panels with separate scales because the deliberately demanding all-nine sensitivity model would otherwise compress the targeted representations; no representation shows a positive improvement that is stable across fold assignments on any outcome. (c) Concurrent whole-item benchmark: median relative held-out loss reduction for the six item-level outcomes, with the entire repeated-split range below zero for every outcome.

4.5. Synthesis. The two sets of results are not in tension with one another. Dense automated facial-inference data contained robust, channel-specific person differentiation across item positions, preserved under a one-effective-tick stress test and 500 model-free item splits, and yet the evaluated aggregate person scores and concurrent whole-item activation features improved none of the eleven specified held-out benchmarks. Within this design, measurement repeatability and criterion utility were separate empirical properties, and evidence for the first did not transfer to the second.

5. Discussion

5.1. The dissociation between repeatability and criterion utility. The central result is a dissociation between within-session repeatability and criterion utility. Eight of nine activation channels order persons almost as consistently under a one-effective-tick stress test as under nominal density ($E[\rho^2] = .881\text{--}.963$), and a model-free split-half design reproduces the pattern; yet none of the evaluated aggregate representations of those same channels improved held-out prediction of five

person-level or six item-level benchmarks. In classical language, the channels behave within this session like highly reliable instruments whose object of measurement is unidentified, while the coarse scores built from them added no criterion information beyond the specified baselines. Neither half of the result should be generalized into the other: high generalizability coefficients do not establish usefulness, and null criterion increments do not show that the channels are noise (the stress-test coefficients directly refute that reading). Reviews of measurement practice have repeatedly warned against treating reliability as a unitary property that transfers across scores, decisions, and uses (Ellis, 2021; Flake & Fried, 2020; Hedge et al., 2018); these data provide an empirical instance in which the two properties dissociate within a single dataset.

It is natural to ask what is being measured so repeatably. The answer supported by the design is a stable within-session person signature in the channel outputs whose source the design cannot identify. Candidate contributors include stable facial morphology and expressivity, person-specific camera and lighting conditions, administration mode and the session conditions correlated with it, session-level algorithm calibration, and any genuinely affect-relevant individual differences, all of which are confounded with the person in a one-session design. The title of this article names the three families: a psychological signal, a transient state, or a sensor style, meaning everything about how a particular person appears to a particular capture pipeline. Two observations locate part of the signature in the third family without excluding the others. The mode-linked composition of the general activation score (Section 4.3) shows that part of the shared variance covaries with session conditions; and the near-zero PC1 loadings of the two project-named channels Anxious and Attentive alongside their .92 mutual correlation suggest structured processing families rather than a psychological dimension. The measurement conclusion is therefore bounded: the channels support repeatable relative person scores within this session and task, whereas the interpretation of those scores is unresolved and cannot be resolved by reliability evidence alone (Borsboom et al., 2004; Campbell & Fiske, 1959).

5.2. Effective information in dense process streams. The accounting has a general implication for how dense sensor streams should be reported. This dataset contains nearly 900,000 aligned ticks, yet under an AR(1) approximation a 217-tick cell carries the information of at most roughly 66 to 143 independent observations (Appendix C explains why the figure is, if anything, an overstatement), and our

headline coefficients credit it with one. The stress test also clarifies what density contributes. Dense ticks did not create person-level information (the coefficients barely move between the AR(1) and nominal scenarios), but they made the measurement structure auditable, since sparse channels, coverage artifacts, and serial dependence are all visible and quantifiable in a dense stream; the mode-linked truncation of the in-person recordings (Section 2.4) is a case in point. We suggest that researchers reporting dense process channels routinely publish an effective-information account (autocorrelation, effective observations per analysis unit, and at least one conservative sensitivity scenario) alongside raw volume, in the same way that sample-size reporting is expected alongside effect sizes (McNeish, 2026; Schneider & Junghaenel, 2023).

5.3. Provenance as a measurement property. Three of the nine channels reached this analysis carrying project-assigned names rather than the software’s own output labels: Anxious and Attentive are renamed outright, and Negative is, in addition to its renaming, a sign-folded transform whose direction is unrecoverable. Such renamings are not unusual in long-lived collaborative datasets, but they are not neutral for interpretation, since a criterion association with a channel named Anxious would suggest a claim about anxiety that the provenance cannot support. The provenance crosswalk (Appendix B) is intended to prevent the most consequential silent inflation of claims in this literature, namely the drift from variable label to construct (Barrett et al., 2019; Flake & Fried, 2020). The same transparency obligation extends to the administration itself: the source publications describe an online study, whereas the archival record documents a mixed-mode design, and the coverage and recording-volume asymmetries of Section 2.4 show that the difference is measurement-relevant, not merely historical. Nothing in the present analyses re-evaluates the substantive conclusions of the source studies, which rest on different questions and models; the administration record matters here because it bears on the measurement properties under study. The absence of independent technical documentation for the specific commercial system likewise requires explicit reporting; benchmark validations of other commercial classifiers (Dupré et al., 2020; Kulke et al., 2020; Stöckli et al., 2018) inform the genre but cannot substitute for system-specific validation, and we make no accuracy claim on the system’s behalf.

5.4. Interpretation of the null criterion increments. The criterion benchmarks were designed so that the coarse aggregate representations were not disadvantaged:

shared folds, training-only standardization, baselines fixed in advance from the historical analysis tables, and outcomes spanning a wide range of baseline performance. The uniformly non-positive median increments, including for math anxiety, the outcome the Anxious channel’s project history most directly targets, are then informative: no evaluated representation improved any of the eleven held-out benchmarks under these learners, baselines, and samples. That statement is bounded in two directions. Cross-validated non-improvement does not establish zero conditional association, and it is a claim about the evaluated aggregate representations rather than about the channels’ latent content. The result is consistent with the finding that complex affect-dynamic summaries often add little beyond simpler information (Dejonckheere et al., 2019), here in a setting where the simpler information is the assessment’s own log data.

Several boundaries qualify the finding. These are exploratory benchmarks on $n = 100\text{--}169$ persons; the representations evaluated are coarse aggregates; the whole-item benchmark tests concurrent summaries, not event-locked or dynamic features; and a negative median ΔR^2 under cross-validation is not an equivalence demonstration. The results therefore do not show that facial process data are useless, that emotion is unrelated to math anxiety or behavior, or that richer representations (within-item trajectories, error-locked windows, cross-modal fusion) would fail; they show that repeatability and density did not, by themselves, yield criterion utility here. Testing richer, leakage-safe representations against these same baselines is a separate undertaking with its own design requirements, which we do not preempt.

5.5. Recommendations for practice. Several reporting practices follow from this workflow for researchers who attach automated affect channels, or any dense sensor stream, to assessment data.

- (1) A provenance crosswalk should be published that maps every analyzed variable to its source field, historical renames, and transformations, and that states each channel’s interpretation ceiling at first mention; the administration and capture conditions of the stream, including mode changes, belong in the same record (Appendices A and B).
- (2) Effective information should be reported alongside raw volume: autocorrelation and effective-observation accounting per analysis unit, with at least one conservative information scenario next to the nominal one (Table 3).
- (3) Reliability evidence should be attached to a score and a decision rather than to the instrument, specifying the measurement component, the facet design,

and the decision type; components of the same channel can be expected to differ (Figure 4b).

- (4) Model-based coefficients should be corroborated model-freely; a repeated split-half on the natural replication facet is inexpensive and diagnoses parametric artifacts.
- (5) Repeatability claims should be kept separate from utility claims. Where utility is claimed, it should be evaluated out-of-sample against the assessment’s own baseline data with shared folds and training-only preprocessing; where it is not evaluated, it should not be implied from reliability.

5.6. Limitations. Six limitations bound the claims. First, the design is a single session of a single fixed-order task, administered in two modes that are completely confounded with calendar time; item content and serial position are confounded, cross-session stability is unidentified, and the mode-linked differences in coverage and recording volume are part of the session signature that the design cannot decompose. Second, no momentary self-report, human facial coding, or independent physiological channel is aligned to the stream, so no analysis here validates any channel as a measure of subjective emotion; our conclusions concern repeatability and incremental utility, and stop short of construct validity. Third, the commercial system’s internal processing is undocumented for this administration, so software-side processing families remain a rival explanation that the design cannot exclude. Fourth, the criterion benchmarks are exploratory, with modest samples ($n = 100$ for three outcomes) and coarse representations; they bound those representations, not the modality. Fifth, the generalizability coefficients are point estimates on a latent scale, conditional on the realized coverage pattern; we did not propagate estimation uncertainty into them. Sixth, missing cells and sparse channels (Joy especially) are handled by explicit support policies rather than imputation, so conclusions about the sparsest components rest on flagged, reduced support.

5.7. Future directions. Three extensions appear worth pursuing, and each requires design elements that this archive lacks. Event-locked analyses, for instance of channel behavior in windows around first errors, need validated event timestamps and leakage-safe feature construction. Construct validation needs concurrent criteria (momentary self-report, blinded human coding, or additional modalities) aligned at the same person-item-time grain (Bian et al., 2018; Horvers et al., 2024; Kuklick et al.,

2026). Transportability needs repeated sessions, randomized item orders, and administration modes crossed with persons rather than confounded with them, so that person, session, content, and mode can be separated. Until such data exist, the present results establish which components of which channels replicate, under what information accounting, and with what interpretation ceiling.

6. Concluding remarks

Dense automated facial-inference recording during a cognitive assessment produced channel outputs whose person ordering was strongly repeatable across item positions, in the sense that it was preserved under a one-effective-tick stress test and under model-free item splits, and whose evaluated aggregate representations nonetheless improved no specified held-out criterion benchmark. We conclude that observation density, measurement repeatability, and criterion utility are three distinct properties and that dense process data should be credited with each only on its own evidence. The value of such streams for psychological measurement will be decided not by the volume of recording but by explicit statement of the measurement objects, transparent accounting of the information they contain and the conditions under which they were captured, and demonstrated utility against the data that an assessment already collects.

Software and reproducibility. All analyses were conducted in R (R Core Team, 2026); the mixed models were fitted with `lme4` (Bates et al., 2015). Reproducibility is documented at three distinct levels in Appendix G (release consistency, a documented consumer and quality-assurance environment snapshot, and the unrecorded historical fitting runtime). A replication package containing selected read-only source modules for the reported evidence chain, frozen aggregate result releases, verification of the contracted artifact classes and headline quantities, and a machine-readable claim–evidence matrix is available at [TO BE ADDED: repository URL].

Data availability. The archival dataset contains facial-inference recordings and is subject to participant-privacy constraints that permit no redistribution and provide no access route for third parties. The replication package therefore shares the selected evidence-chain modules together with a disclosure-screened aggregate layer. Its tables have several nonparticipant grains, including channel, channel-by-component-by-scenario, outcome-by-model, repetition-by-fold, and model-rank audit rows; this list describes examples rather than an exhaustive schema. No released object is at the

grain of a participant or of a participant-by-item cell. A 35-target contract classifies manuscript artifacts as regenerated, verified static, public-number verified, partially withheld, or withheld; the runnable exhibit layer rebuilds ten table files and four public figures. The withheld participant-dependent figure and figure panel, and every other exception, are recorded with their reason and aggregate substitute. Superseded branches, input-bound configuration, cell summaries, and person scores do not ship.

Ethics and consent. The present study is a secondary analysis of de-identified archival research data. Ethics approval and consent procedures for the original data collection are described in the source publications (Wang et al., 2021, 2025). No identifiable individual data, images, or videos are reported; only nonparticipant aggregate outputs are released, while the underlying analyses used the restricted archival data. The analyses reported here are exploratory secondary analyses and were not preregistered.

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Online Supplement

Signal, State, or Sensor Style? Generalizability and Interpretive Limits of Automated Facial-Inference Channels During Cognitive Assessment

JoonHo Lee and Yurou Wang

This supplement accompanies the main text. Section, equation, table, and figure numbers below are prefixed by the appendix letter (A.1, B.1, ...). References to the main text use its unprefix numbering.

Contents

Appendix A. Participant flow, administration modes, and observation accounting	39
A.1. The cohort accounting as a reconciliation	39
A.2. Administration modes and collection waves	40
A.3. Person-by-item cell coverage	41
A.4. Tick-level observation accounting	42
A.5. Criterion-analysis availability	42
Appendix B. Channel provenance, historical transform, and terminology contract	42
B.1. Full provenance crosswalk	42
B.2. Historical transform and its identifiability consequences	44
B.3. Terminology contract	44
Appendix C. Generalizability models: specification, effective information, and full tables	45
C.1. Model specification	45
C.2. Effective information under serial dependence	46
C.3. Coefficient definition	47
C.4. Full activation results	47
Appendix D. Repeated item split-halves: procedure, support policy, and full results	48
D.1. Procedure	48
D.2. Sparse-support reporting policy	49

D.3.	Full results for all three components	49
D.4.	Reading the component heterogeneity	49
Appendix E.	Cross-channel structure, recording covariates, and administration-mode descriptives	50
E.1.	Person-score construction and raw-score agreement	50
E.2.	Correlations among activation person scores	51
E.3.	Principal components	51
E.4.	The recording-volume rival explanation	51
E.5.	Administration-mode descriptives	51
Appendix F.	Criterion benchmarks: specifications and complete results	54
F.1.	Person-level benchmark: model roster	54
F.2.	The mathematics self-report coding correction	54
F.3.	The task-covariate correction	55
F.4.	Person-level benchmark: complete results	56
F.5.	In-sample descriptive supplement	56
F.6.	Item-level benchmark: specification and complete results	56
Appendix G.	External review, statistical quality assurance, claim contract, and reproducibility	59
G.1.	External code-and-claims review	59
G.2.	Second external review and the task-covariate correction	59
G.3.	Statistical-content quality assurance	60
G.4.	Claim–evidence contract	60
G.5.	Reproducibility and environment	61
G.6.	Administration-mode descriptives: derivation and verification	62

Appendix A. Participant flow, administration modes, and observation accounting

A.1. The cohort accounting as a reconciliation. The archive assembles three separately stored sources: the participant-level survey table underlying the published analyses (Wang et al., 2021), the combined cleaned test logs, and the facial-inference database. Each source has its own cohort, and no single attrition funnel generates all of them (for example, eight test-log participants are absent from the survey table, and six survey participants are absent from the test logs). [Table A.1](#) therefore reports

the audited cohort reconciliation. All counts were verified against the stored archival files by the reconstruction audit; identifiers themselves are never released.

TABLE A.1. Cohort Reconciliation Across the Three Archival Sources

Cohort	n	Definition
Survey (published)	177	Participant-level survey/analysis table of the 2021 publication
Test logs	179	Participants in the combined cleaned test logs (4,478 rows: 4,475 canonical item 1–25 rows plus three historical item-26 artifact rows)
Facial database, cleaned	176	After the historical rule removing every session of duplicated facial-database identifiers
Matched cohort	169	Test-log $\cap$ cleaned facial cohort whose clocks enter the historical time-matching table; <i>analytic cohort for all measurement analyses</i>
Three-source overlap	162	Participants linkable across survey, test, and facial sources
Mathematics-task subset	111	Survey-table subset with the mathematics task (the 2025 publication sample)

Note. Selected set differences, verified from actual identifier comparisons (aggregate counts only): test logs minus survey = 8; survey minus test logs = 6; test logs minus cleaned facial cohort = 10 (these 10 participants account for 250 canonical test items with no facial summary); cleaned facial cohort minus test logs = 7; survey minus matched cohort = 15; matched cohort minus survey = 7.

A.2. Administration modes and collection waves. The survey records an explicit mode variable: of the 177 surveyed participants, 47 are marked **on site** and 130 **online**. The facial-database session timestamps partition into two non-overlapping waves, February 18 to March 13, 2020 and September 2, 2020 to January 29, 2021, with no sessions in between, and the partition coincides exactly with the survey variable: every early-wave participant with a survey record is marked on site, and every late-wave participant with a survey record is marked online. The last in-person session (March 13, 2020) is the date of the university’s COVID-19 campus closure. Within the 169-person matched cohort, 42 participants belong to the in-person wave and 127 to the online wave; the 7 matched participants without survey records all have sessions in the online-wave window, so their mode designation rests on the timestamp partition alone. The source publications (Wang et al., 2021, 2025) describe the procedure as administered online over Zoom; the archival record documents the mixed-mode history reported here, and the main text (Section 2.1) describes the administration as the archive records it.

Mode is completely confounded with calendar time, and with every session-level condition that changed between the waves (camera path, physical setting, any software or configuration drift). For the online sessions the software read the participant’s face from the Zoom window on the study computer (Wang et al., 2021), so the online stream was computed from a compression-mediated, re-rendered video window fed by the participants’ own webcams; the archival record does not document the in-person capture path (the vendor documentation lists USB, built-in, and network cameras as supported inputs, and direct capture is the natural configuration, but this cannot be verified from the stored files). The realized tick rate is essentially identical in the two modes (median per-cell density 5.79 vs. 5.76 Hz), so the software’s output cadence did not differ; nothing follows from this about the quality of the video input, which is undocumented in both modes. All mode-related quantities in this article are archival descriptives derived from the immutable stores (Section G.6); none is interpreted as an effect of mode.

One further historical variable is documented here because it appears in the stored files: the test logs carry an internal label (`subject`, levels `Numeric` and `Logic`) that reflects the data-collection path and file lineage. The entire in-person subgroup and the online literacy-task group are stored under `Logic`; the online mathematics-task group is stored under `Numeric`. The label therefore does not coincide with task version, mode, or any single design factor, and no analysis interprets it; the validated survey task variable carries all task information in the criterion benchmarks (Section F.3).

A.3. Person-by-item cell coverage. For the 169-person matched cohort, $169 \times 25 = 4,225$ person-by-item cells are possible and 4,126 (97.7%) contain an aligned facial-inference summary. The 99 missing cells are not scattered at random: 97 of the 99 belong to in-person participants, the cells concentrate in the final item positions (of the 99: item 25, 42 cells; item 24, 26; item 23, 15), and 41 of the 42 in-person participants have at least one unobserved cell, a pattern consistent with facial recordings that ended before the participant finished the last items. Online coverage is nearly complete (3,173 of 3,175 possible cells; the two missing online cells are single cells at items 5 and 25 for two participants). In-person coverage is 953 of 1,050 possible cells (90.8%). Missing cells are treated as absent observation opportunities throughout; they are never imputed and never entered as zeros.

A.4. Tick-level observation accounting. The facial-inference database contains 964,143 time-stamped ticks for the matched cohort, of which 896,309 fall inside a person’s item intervals after the historical clock correction and are assigned to person-by-item cells; ticks outside item intervals contribute only to session-level recording covariates ([Appendix E](#)). Observed cells contain a mean of 217.2 and a median of 153 eligible ticks. By mode, the in-person wave contributes 241,119 aligned ticks over 953 cells (median 191 ticks per cell; median item duration 34 s) and the online wave 655,190 ticks over 3,173 cells (median 142; median duration 25 s): in-person items simply ran longer, at the same capture rate. The empirical median interval between successive ticks is approximately 160 ms (about 6 Hz); the 50-ms figure quoted in the source publication describes nominal device capability, not the realized sampling rate of this administration. Because the within-cell series are strongly serially dependent ([Appendix C](#)), none of these tick counts is treated as a count of independent observations anywhere in the analysis.

A.5. Criterion-analysis availability. The person-level criterion benchmark uses complete cases per outcome: $n = 169$ for the two log-derived outcomes (first-attempt rate, mean log duration). For the survey-dependent outcomes, 103 matched participants have math-anxiety scores and 107 have post-error behavioral outcomes; complete cases across outcome and all baseline covariates yield $n = 100$ for math anxiety and both post-error outcomes. The item-level benchmark uses the 4,126 observed cells (1,722 post-error cells for the two post-error outcomes).

Appendix B. Channel provenance, historical transform, and terminology contract

B.1. Full provenance crosswalk. [Table B.1](#) records, for each of the nine project channels, the software output field it derives from, the historical naming steps, the required first-mention wording, and the interpretation ceiling. The crosswalk is stored as a machine-readable configuration file in the replication package and is checked by the quality-assurance pipeline ([Appendix G](#)); channel names in every analysis artifact are bound to it. The main text deliberately develops the argument in terms of the final project channel names alone; this appendix is the authoritative record of where those names came from.

The *Anxious* and *Attentive* names originate in the original research team’s documented analysis decisions (the working anxiety-relevant and attention-relevant channels for the substantive studies); the historical intermediate name **Danger** for the

TABLE B.1. Complete Provenance Crosswalk for the Nine Project Channels

Channel	Source field	Required first-mention wording	Interpretation ceiling
Anxious	Surprise	project-derived automated facial-inference Anxious channel (vendor source field: Surprise)	Not a vendor anxiety field, self-reported anxiety, clinical anxiety, or criterion-validated momentary anxiety
Attentive	Engagement	project-derived automated facial-inference Attentive channel (vendor source field: Engagement; historical intermediate name: Danger)	Not a direct measure of attention, task engagement, vigilance, or danger perception
Negative	Valence	project-derived Negative channel (sign-folded absolute magnitude of vendor Valence; valence direction not retained)	Not negative affect, negative-emotion probability, unpleasantness, or signed valence
Joy	JOY	automated facial-inference Joy channel (vendor source field: JOY)	Not self-reported or criterion-validated subjective joy
Sad	SAD	automated facial-inference Sad channel (vendor source field: SAD)	Not self-reported or criterion-validated subjective sadness
Anger	Anger	automated facial-inference Anger channel (vendor source field: Anger)	Not self-reported or criterion-validated subjective anger
Fear	Fear	automated facial-inference Fear channel (vendor source field: Fear)	Not self-reported or criterion-validated subjective fear
Disgust	Disgust	automated facial-inference Disgust channel (vendor source field: Disgust)	Not self-reported or criterion-validated subjective disgust
Contempt	Contempt	automated facial-inference Contempt channel (vendor source field: Contempt)	Not self-reported or criterion-validated subjective contempt

Note. All nine channels are outputs of the same vendor pipeline recorded under the same session conditions; shared vendor processing is therefore a standing rival explanation for any cross-channel commonality.

Engagement-derived channel is preserved in the crosswalk because it appears in intermediate archival files. These naming decisions are treated as fixed provenance and are reported, not relitigated.

B.2. Historical transform and its identifiability consequences. For every channel, the historical pipeline stored the transformed value

$$H = \begin{cases} 0, & |X| = 0, \\ \log |X|, & |X| > 0, \end{cases} \quad (\text{B.1})$$

where X is the vendor source value. Three consequences constrain all downstream analysis. First, source sign is destroyed; this matters only for **Valence**, whose sign carried the substantive direction, so the project *Negative* channel is an absolute-magnitude quantity and no analysis of it can recover pleasant–unpleasant direction. Second, $|X| = 1$ maps to $H = 0$, the same stored value as $X = 0$; activation therefore cannot be reconstructed from H alone and is defined from raw-source nonzero status in the archival stream. Third, magnitudes are analyzed on the historical $\log |X|$ scale; an active tick can legitimately have $H = 0$ (from $|X| = 1$), which is why the hurdle decomposition distinguishes the *active boundary* (whether the transformed magnitude exceeds its floor among active ticks) from the *strict-positive magnitude*.

B.3. Terminology contract. The manuscript’s language is bound by a terminology contract fixed before drafting. Terms in the left column are the defaults; terms in the right column are not used unless separately validated evidence exists (none does in this archive).

Six boundaries follow from the design and are maintained in every section of the main text: (1) no subjective emotion is identified; (2) no causal effect is identified, including of administration mode; (3) no cross-session trait is identified (all stability statements are within-session); (4) no event-locked dynamics were tested; (5) no universal-nonutility conclusion is drawn (null increments are bounded to the evaluated representations and benchmarks); and (6) non-significance is never read as equivalence.

TABLE B.2. Terminology Contract

Used in this article	Avoided (not licensed by this design)
project-derived automated facial-inference channel	AI-measured anxiety; true joy; emotion ground truth
channel activation; active magnitude	the participant felt X; emotion occurred
stable person/session signature	trait emotion; anxious person; attentive person
person differentiation across item positions	cross-session trait
item/position-linked variation	pure item effect; item-induced emotion
recording opportunity; observability	unobserved ticks as zeros
administration mode (archival descriptive)	mode effect; Zoom effect; in-person effect
evaluated representation; specified benchmark	facial data (in general) do not predict
no incremental improvement for the evaluated representations	equivalence; facial features worsened behavior; universal nonutility
896,309 aligned ticks (serially dependent)	896,309 independent observations
descriptive cross-channel commonality (PC1)	general emotion factor; negative-affect factor; latent distress

Appendix C. Generalizability models: specification, effective information, and full tables

C.1. Model specification. For each channel k , the activation model is a binomial generalized linear mixed model fitted to the cell-level counts of the 169-person cohort with `lme4` (Bates et al., 2015),

$$\text{active}_{pi} \sim \text{Binomial}(m_{pi}, \pi_{pi}), \quad \text{logit}(\pi_{pi}) = \beta_i + u_p + w_{pi}, \quad (\text{C.1})$$

where m_{pi} is the number of eligible ticks in cell (p, i) , β_i ($i = 1, \dots, 25$) are item-position fixed effects, $u_p \sim N(0, \sigma_p^2)$ is the person effect, and $w_{pi} \sim N(0, \sigma_{pi}^2)$ is the person-by-item effect. Models were estimated by Laplace approximation with an extended optimizer budget. On the logit scale, the tick-level residual is the standard-logistic variance $\pi^2/3 \approx 3.290$ (Nakagawa & Schielzeth, 2010; Nakagawa et al., 2017). Secondary components use the analogous specification: a binomial model for the active boundary among active ticks, and a Gaussian linear mixed model (restricted maximum likelihood) for strict-positive magnitudes on the historical $\log |X|$ scale.

Because item positions enter as fixed effects, the fitted model conditions on the observed 25-position form; the decision-study coefficient below treats the person-by-item and residual components as the error entailed by a hypothetical exchangeable item/position universe of size 25. This “random item/position” target is a model-based idealization declared in the analysis configuration (the observed form was not sampled from any item bank), and all coefficients are conditional on the realized coverage pattern. The models pool the two administration modes without a mode term; mode-linked level differences are therefore absorbed into σ_p^2 , which is one reason the person component is interpreted as a person-plus-session signature rather than a person trait (Section 5.1).

One estimation note is required by the reporting policy: the Joy activation fit was optimizer-sensitive (a documented sensitivity refit is part of the frozen release), so its exact model-based coefficient should not be interpreted precisely. The model-free split-half evidence (Appendix D) is the corroborating source for Joy’s lower stability, and no conclusion in the main text depends on Joy’s precise coefficient value.

C.2. Effective information under serial dependence. For a weakly stationary within-cell series with residual variance σ_r^2 and autocorrelations ρ_ℓ , the variance of the cell mean of m ticks is

$$\text{Var}(\bar{r}) = \frac{\sigma_r^2}{m} \left[1 + 2 \sum_{\ell=1}^{m-1} \left(1 - \frac{\ell}{m} \right) \rho_\ell \right] = \frac{\sigma_r^2}{m_{\text{eff}}}, \quad (\text{C.2})$$

which defines the effective tick count m_{eff} . Under a lag-1 autoregressive structure with parameter ρ_1 and large m , this reduces to the familiar

$$m_{\text{eff}} = m \frac{1 - \rho_1}{1 + \rho_1}. \quad (\text{C.3})$$

The release computes Equation (C.3) with $m = 217.2$ (the mean eligible tick count per observed cell) and $\hat{\rho}_1$ the channel’s median lag-1 autocorrelation across cells with nonconstant series. Three features of this construction all point in the same direction. Dependence beyond lag 1 is ignored; the mean count overweights large cells relative to the harmonic-type aggregate that governs a person’s item-average error; and lag-1 autocorrelations of sparse binary series understate the latent dependence, most severely for the sparsest channels. The channel-level m_{eff} values are therefore best read as upper bounds on typical within-cell information, which is what motivates the one-effective-tick stress test ($m_{\text{eff}} = 1$) as the headline scenario. The stress test

is conservative but is not a formal lower bound: it is a transparent worst-case-style scenario for within-cell information, not a bound derived from the data.

C.3. Coefficient definition. The relative (person-ordering) generalizability coefficient for the 25-item-position design is

$$E[\rho^2] = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_\delta^2}, \quad \sigma_\delta^2 = \frac{\sigma_{pi}^2}{25} + \frac{\pi^2/3}{25 m_{\text{eff}}}, \quad (\text{C.4})$$

i.e., person-by-item interaction and tick-level residual are relative error, averaged over 25 item/positions, with the residual scaled by the effective tick count of the scenario. We do not report the absolute-decision coefficient Φ : an absolute coefficient charges item/position main effects to error, which is meaningful only when absolute score levels carry validated interpretations, and no absolute-level interpretation of these channels is supported by the design ([Appendix B](#)). All coefficients are point estimates on the component-model latent scale; estimation uncertainty is not propagated into them, which is stated as a limitation in the main text.

C.4. Full activation results. [Table C.1](#) reports, for each channel, the latent-scale variance components, the median lag-1 autocorrelation, the AR(1) effective tick count, and $E[\rho^2]$ under all three scenarios.

The variance components account for the main-text pattern. For the primary eight channels, σ_p^2 is comparable to or larger than σ_{pi}^2 , so averaging over 25 item/positions leaves person ordering nearly intact even when each cell contributes a single effective observation. For Joy, σ_{pi}^2 exceeds σ_p^2 by a factor of almost fifty, so additional within-cell information cannot restore cross-position consistency; the instability lies between positions rather than within them.

TABLE C.1. Activation Generalizability: Variance Components and Coefficients Under Three Effective-Information Scenarios

Channel	σ_p^2	σ_{pi}^2	Lag-1	AR(1)	m_{eff}	$E[\rho^2]$ by scenario		
						One tick	AR(1)	Nominal
Anxious	3.203	1.505	.520		68.6	.943	.981	.981
Attentive	2.080	1.429	.512		70.2	.917	.972	.973
Negative	1.418	1.506	.532		66.4	.881	.958	.959
Joy	0.764	36.763	.329		109.7	.323	.342	.342
Sad	3.135	2.648	.373		99.3	.930	.967	.967
Anger	2.427	2.487	.329		109.6	.913	.960	.960
Fear	5.292	1.807	.206		143.1	.963	.986	.986
Disgust	1.624	1.267	.352		104.2	.899	.969	.969
Contempt	2.581	3.148	.495		73.4	.909	.953	.953

Note. Latent (logit) scale; tick-level residual $\pi^2/3 \approx 3.290$ for every channel; mean 217.2 eligible ticks per observed cell. “Lag-1” is the median lag-1 autocorrelation of within-cell activation series (cells with nonconstant series). Scenarios as defined in [Section C.2](#); coefficients from [Equation \(C.4\)](#). All values are point estimates, conditional on the observed 25-item-position task and realized coverage pattern; the exchangeable item/position universe is model-based; no cross-session claim is made. Joy’s dominant σ_{pi}^2 reflects extreme sparsity (nonconstant activation series in only 494 of 4,126 cells) and an optimizer-sensitive fit; its exact coefficient should not be interpreted precisely.

Appendix D. Repeated item split-halves: procedure, support policy, and full results

D.1. Procedure. The split-half design is model-free and operates on cell-level component values (activation proportion, boundary proportion among active ticks, mean strict-positive magnitude). In each of 500 replications: (1) the 25 item positions were randomly permuted and partitioned into halves of 12 and 13 positions; (2) each participant’s half-scores were the means of their observed cell values within each half, requiring at least three observed cells per half; and (3) the replication statistic was the Spearman rank correlation between half-scores across all participants with valid scores in both halves, computed only when at least 20 such participants existed. The Spearman–Brown value $2r/(1+r)$ is reported as a heuristic on two counts: the halves are unequal (12 vs. 13 items), so the classical equal-halves correction does not hold exactly, and the correction is derived for Pearson correlations of parallel composites under classical test theory rather than for rank correlations. Half-scores are raw means over observed cells and deliberately involve no item/position adjustment (the model arm of the analysis carries that adjustment), so position main effects enter person ordering through each participant’s realized coverage mix; this contribution is

bounded by the 2.3% missing-cell share and, per [Section A.3](#), concentrated in the in-person group. Summaries are medians across the 500 replications with 2.5th–97.5th percentile ranges; these ranges describe stability across random item splits and are not inferential confidence intervals.

D.2. Sparse-support reporting policy. Adopted after external review: for the two secondary components, any channel whose median supported sample across replications falls below 40 participants is labeled VERY SPARSE SUPPORT; its raw statistic, support counts, and finite-statistic proportion are reported, but its Spearman–Brown value and range are suppressed from headline reporting. The threshold is a transparency flag equal to twice the 20-person computation minimum; it is a descriptive reporting rule rather than an identification criterion, an inference criterion, or a judgment of the channel. Exactly two of the 27 channel-component combinations fall below it: Joy’s active boundary (median $n = 24$; finite statistic in 494/500 replications) and Joy’s strict-positive magnitude (median $n = 20$; finite in 285/500).

D.3. Full results for all three components. [Table D.1](#) lists all 27 channel-component combinations.

D.4. Reading the component heterogeneity. Three observations structure the main-text interpretation. First, activation is the uniformly best-supported component: full support (169) for every channel and adjusted medians of .703–.982. Second, the secondary components are heterogeneous in a channel-specific way: Anxious and Attentive remain high on both secondary components, Disgust and Contempt are intermediate, whereas Negative, Sad, Anger, and Fear drop sharply on the boundary component even where magnitude is moderate. A channel’s repeatability evidence therefore attaches to a specific scored component rather than to the channel name. Third, Joy’s secondary components are cases of sparse support rather than estimated instability: with median supported samples of 24 and 20, and only 57% of magnitude replications even computable, the appropriate summary is that the design cannot support headline stability claims for these components. This is different from claiming that the components are unstable, and different again from claiming that the channel is malfunctioning.

TABLE D.1. Repeated Unequal (12/13) Item Split-Half Stability: All Channels and Components

Component	Channel	Median n	Raw half ρ	Adjusted	2.5%–97.5% range	Finite
Activation	Anxious	169	.964	.982	[.971, .987]	500/500
	Attentive	169	.943	.971	[.956, .980]	500/500
	Negative	169	.879	.936	[.907, .951]	500/500
	Joy	169	.542	.703	[.633, .757]	500/500
	Sad	169	.774	.873	[.847, .892]	500/500
	Anger	169	.765	.867	[.835, .890]	500/500
	Fear	169	.861	.925	[.910, .942]	500/500
	Disgust	169	.887	.940	[.920, .953]	500/500
	Contempt	169	.815	.898	[.865, .924]	500/500
Active boundary	Anxious	169	.916	.956	[.938, .970]	500/500
	Attentive	169	.790	.882	[.852, .909]	500/500
	Negative	166	.259	.411	[.289, .519]	500/500
	Joy	24	–.047	—	—	494/500
	Sad	116	.329	.496	[.308, .621]	500/500
	Anger	124	.134	.237	[–.018, .410]	500/500
	Fear	95	.157	.272	[.010, .469]	500/500
	Disgust	164	.600	.750	[.688, .809]	500/500
	Contempt	140	.549	.709	[.640, .773]	500/500
Strict-positive magnitude	Anxious	167	.907	.951	[.932, .964]	500/500
	Attentive	169	.892	.943	[.921, .957]	500/500
	Negative	166	.684	.812	[.758, .853]	500/500
	Joy	20	.337	—	—	285/500
	Sad	101	.426	.598	[.455, .713]	500/500
	Anger	112	.382	.553	[.418, .664]	500/500
	Fear	84	.424	.596	[.423, .722]	500/500
	Disgust	154	.689	.816	[.769, .857]	500/500
	Contempt	130	.344	.512	[.375, .628]	500/500

Note. Medians across 500 replications. “Adjusted” is the Spearman–Brown heuristic $2r/(1+r)$; ranges are 2.5th–97.5th percentiles of the adjusted statistic across replications (empirical stability ranges, not confidence intervals). “—” = suppressed under the VERY SPARSE SUPPORT policy (Section D.2); the raw statistic and support counts remain reported. “Finite” = replications with a computable statistic out of 500 attempted.

Appendix E. Cross-channel structure, recording covariates, and administration-mode descriptives

E.1. Person-score construction and raw-score agreement. Person-level activation scores are the estimated person random intercepts (conditional modes) from the activation models of Appendix C, one score per channel per participant ($n = 169$). To confirm that no conclusion depends on model-based scoring, the release also computes

raw person scores (each participant’s mean cell-level activation) and their agreement with the model scores: for activation, Spearman correlations between raw and model scores are .931–.998 across the nine channels (Pearson .698–.984, the lower Pearson values reflecting shrinkage-induced nonlinearity rather than reordering). Agreement is lower for the sparser secondary components and is not computable for Joy’s active boundary. All structure analyses use the activation model scores.

E.2. Correlations among activation person scores. [Table E.1](#) gives the full correlation matrix underlying main-text Figure 5a.

TABLE E.1. Correlations Among the Nine Activation Person Scores (Lower Triangle)

	Anx.	Att.	Neg.	Joy	Sad	Ang.	Fear	Dis.	Con.
Anxious	1								
Attentive	.925	1							
Negative	−.004	.199	1						
Joy	−.030	.016	.370	1					
Sad	−.300	−.115	.674	.186	1				
Anger	−.251	−.019	.720	.352	.744	1			
Fear	.281	.221	.163	.280	.052	.138	1		
Disgust	.223	.362	.414	.229	.360	.409	.068	1	
Contempt	.002	.142	.635	.303	.502	.648	.115	.281	1

Note. $n = 169$. The structure is as follows: an Anger–Sad–Negative–Contempt cluster ($r = .502$ –.744), a highly correlated Anxious–Attentive pair ($r = .925$; their common provenance family is documented in [Appendix B](#)), weak-to-negative correlations between the two groups, and a Joy score modestly related to the first cluster and essentially unrelated to the second.

E.3. Principal components. [Table E.2](#) reports the full variance decomposition and the PC1 loadings.

E.4. The recording-volume rival explanation. [Table E.3](#) reports both correlation types for the three session-level recording covariates.

E.5. Administration-mode descriptives. Because administration mode is a session-level condition bundled into the same rival-explanation family, [Table E.4](#) reports the mode composition of the general activation score and of the recording-volume covariate, and [Table E.5](#) reports the per-channel picture. The in-person group sits higher on the general score and on recording volume, but the channel-level pattern is heterogeneous: the standardized mode differences are largest for Anxious, Attentive, Fear, and Disgust ($d \approx 0.6$ –0.7) and near zero for Sad, Anger, and

TABLE E.2. Principal Components of the Standardized Activation Person Scores

Component	Proportion	Cumulative	Channel	PC1 loading
PC1	.384	.384	Anxious	−.029
PC2	.248	.631	Attentive	.087
PC3	.119	.751	Negative	.473
PC4	.081	.832	Joy	.268
PC5	.071	.903	Sad	.428
PC6	.043	.946	Anger	.478
PC7	.026	.972	Fear	.130
PC8	.024	.996	Disgust	.308
PC9	.004	1.000	Contempt	.416

Note. Principal component analysis of the nine standardized activation person scores ($n = 169$); PC1 sign-aligned so its loading sum is positive. PC1 is used descriptively (and as the “general activation score” in the criterion benchmark); it is not interpreted as a general emotion factor or negative-affect factor. The descriptive label licensed by the reporting policy is: *Anger/Sad/Contempt and project-derived Negative-dominant activation signature*, where Negative is the sign-folded Valence-derived project channel ([Appendix B](#)).

TABLE E.3. Associations Between the General Activation Score (PC1) and Session-Level Recording Covariates

Recording covariate	Pearson r	Spearman r
Coverage proportion (observed / possible cells)	−.065	−.164
Log total recorded ticks	.323	.331
Unassigned-tick proportion	.026	.076

Note. $n = 169$. The log-tick association shows that the common score carries recording-volume information; the training-fold-only nuisance adjustment in the criterion benchmark ([Appendix F](#)) bounds the consequence of this rival explanation for the utility conclusion (largest change in any median ΔR^2 : .003). The adjustment does not remove the rival from the structural interpretation: shared vendor processing, expressivity, and camera/session conditions remain unseparated in a one-session design (Podsakoff et al., 2003).

Contempt, so the mode association is not confined to the recording-volume/PC1 component. All of these are archival descriptives of a fully confounded design variable ([Section A.2](#)); they are reported to locate part of the shared variance in session conditions, and no mode effect is claimed. The derivation and verification of these numbers are described in [Section G.6](#).

[Table E.5](#) also bounds the consequence of the pooled-mode design for the repeatability claims. Treating each channel’s score-scale mode share η^2 as if it were a share of the latent person variance σ_p^2 , and removing it, lowers the one-effective-tick coefficient of Equation ([C.4](#)) by at most .007 across the primary eight channels (.011 for

the unstable sparse Joy fit). The approximation is crude in the conservative direction (conditional-mode scores are shrunk, so their between-mode share, applied to the full latent variance, if anything overstates the adjustment), and it uses no refitting; a mode-fixed-effect refit remains the definitive check and is left to future work on these archives.

TABLE E.4. General Activation Score and Recording Volume by Administration Mode (Archival Descriptives)

	In person ($n = 42$)	Online ($n = 127$)
PC1 mean (SD)	0.54 (1.79)	−0.18 (1.85)
Median log total recorded ticks	8.75	8.52
Median eligible ticks per observed cell	191	142
Median item duration (s)	34	25
Median per-cell tick rate (Hz)	5.79	5.76

Note. A Wilcoxon rank-sum test of the PC1 mode contrast gives $p = .013$. Mode is completely confounded with calendar time, camera path, and every session-level condition that changed between the waves; the contrast is therefore a difference between confounded groups, not an effect of mode, and the main text treats it as one strand of the session-condition rival-explanation family (Section 4.3).

TABLE E.5. Activation Person Scores by Administration Mode, per Channel (Archival Descriptives)

Channel	In-person M	Online M	d	η^2	One-tick $E[\rho^2]$ change
Anxious	0.85	−0.25	0.67	.077	.004
Attentive	0.68	−0.22	0.68	.080	.007
Negative	0.25	0.01	0.22	.009	.001
Joy	0.56	0.32	0.53	.050	.011
Sad	0.43	0.41	0.01	.000	.000
Anger	0.59	0.31	0.21	.008	.001
Fear	1.35	0.16	0.61	.066	.003
Disgust	0.63	−0.09	0.62	.068	.007
Contempt	0.51	0.26	0.18	.006	.001

Note. Activation person scores are the conditional modes of Section E.1 (latent-scale units); $n = 42$ in person, 127 online. d is the pooled-SD standardized mean difference; η^2 is the score-scale share of variance associated with mode. “One-tick $E[\rho^2]$ change” is the decrease in the one-effective-tick coefficient of Equation (C.4) when $(1 - \eta^2)\sigma_p^2$ replaces σ_p^2 , an unrefitted bound described in the text. All quantities are archival descriptives of a design variable confounded with calendar time and session conditions; none is an effect of mode.

Appendix F. Criterion benchmarks: specifications and complete results

Both benchmarks are exploratory. They evaluate whether specific, coarse facial representations improve held-out prediction beyond specified baselines; they are not preregistered confirmatory tests, and their conclusions are bounded to the evaluated representations and outcomes.

F.1. Person-level benchmark: model roster. [Table F.1](#) specifies every fitted model.

TABLE F.1. Person-Level Benchmark: Outcomes, Baselines, and Facial Representations

Outcome (n)	Baseline predictors
Math anxiety (100)	Task accuracy, mean response time, sex, the three-category mathematics self-report factor (2 dummies), condition
Post-error help rate (100); post-error skip rate (100)	Math anxiety + the math-anxiety baseline above
First-attempt rate (169); mean log duration (169)	Survey task version with its frozen missing-task indicator, coverage proportion, log total ticks

Note. Four facial representations are added to each baseline: (a) General = activation PC1; (b) Specific = PC1 + Anxious-specific residual (Anxious score regressed on PC1); (c) All nine activation scores; (d) General residualized on the three recording covariates, residualization estimated on training folds only. All models are linear regressions on person-level data. Predictors are standardized with training-fold statistics; folds are drawn at the participant level (five-fold), shared by all models within a repetition; 100 repetitions per outcome. Performance is held-out R^2 per repetition (pooled out-of-fold predictions). For first-attempt rate and mean log duration the baseline already contains the two recording-volume covariates, so those two benchmarks control recording volume by construction. The mathematics self-report item is the archival three-category question (labels *no* / *yes* / *it depends*). The PC1 loadings and the Anxious-specific residualization are outcome-blind transforms estimated once on the full cohort before cross-validation (a disclosed boundary; see main-text Section 3.6).

F.2. The mathematics self-report coding correction. External review found a historical path on which the three categories of the archival mathematics self-report item (stored as 0, 1, 2) could be treated as a continuous score in some criterion baselines. The corrected contract fixes the variable as a nominal factor (0 = *no*, the reference; 1 = *yes*; 2 = *it depends*), with both dummy columns retained in every training fold even when a category is absent from that fold. All five outcomes were recomputed under one versioned pipeline with this contract. [Table F.2](#) shows the

effect on baseline performance: the correction materially strengthened the math-anxiety baseline (median held-out R^2 from .026 to .134). The direction of every facial increment was unchanged: no representation became positive after the correction, so the correction changed baseline strength, not the utility conclusion.

TABLE F.2. Baseline Held-Out R^2 Before and After the Self-Report Factor Correction (Values From the Pre-Task-Correction Pipeline in Which the Comparison Was Run)

Outcome	Historical (continuous path)	Corrected (nominal factor)	Change
First-attempt rate	.0174	.0174	.0000
Math anxiety	.0256	.1339	+.1083
Mean log duration	.7733	.7733	.0000
Post-error help rate	−.0130	−.0318	−.0188
Post-error skip rate	.6991	.6923	−.0068

Note. These comparison values predate the task-covariate correction of [Section F.3](#); the definitive primary-policy results are those of [Table F.3](#).

F.3. The task-covariate correction. A second, independent pre-submission external review found that the frozen criterion releases had violated the task-variable field contract: the item-level benchmark encoded the historical data-collection file label directly, and the person-level builder silently substituted that label for seven participants whose survey task value is missing. The contract is explicit that the historical label, which disagrees with the validated survey assignment for 35 of the 162 participants with known task (800 of 3,952 known-task item rows), must never stand in for the survey task version.

All eleven benchmarks were therefore rerun in a versioned correction release under the contracted survey variable, reusing the original algorithms, repetition counts, folds, and seeds, with a frozen missingness policy decided before the rerun: the *primary* policy retains all participants and adds an explicit missing-task indicator (preserving the full analysis roster of 169 persons and 4,126 cells fixed in the frozen plan); a *complete-case* sensitivity drops the seven missing-task participants (162 persons; 3,952 ordinary and 1,645 post-error cells) instead. The three survey-outcome benchmarks are unchanged by construction (their $n = 100$ mathematics-task subsample contains no missing-task participant and their baselines contain no task term).

Every conclusion is preserved under both policies: all 40 person-level increment medians (20 per policy) and all twelve item-level medians (six per policy) remain negative, no probability-of-improvement exceeds .29 anywhere, and the largest median

shift relative to the superseded frozen values is .005. The visible changes are that the corrected first-attempt-rate baseline is weaker than the label-based one (median held-out R^2 $-.020$ vs. the superseded .017; complete-case .006), and the item-level medians move by at most .004. All main-text criterion tables report the corrected primary-policy values; the complete-case person-level values for the two affected outcomes are: first-attempt rate ($n = 162$): baseline .006, General $-.011$, Specific $-.015$, All nine $-.074$, Nuisance-adjusted $-.013$; mean log duration ($n = 162$): baseline .760, General $-.007$, Specific $-.010$, All nine $-.031$, Nuisance-adjusted $-.003$.

F.4. Person-level benchmark: complete results. Table F.3 reports, for every outcome and representation, the median held-out statistic with its 2.5th–97.5th percentile repeated-split range and the proportion of repetitions in which the representation improved held-out R^2 .

The largest absolute change produced by the nuisance adjustment in any outcome’s median ΔR^2 is .003, so the recording-volume adjustment neither improves nor reverses any conclusion.

F.5. In-sample descriptive supplement. For completeness, the release also reports in-sample standardized coefficients of the two registered facial scores (general, Anxious-specific) in each outcome’s full-data model, with nonparametric bootstrap percentile intervals (1,000 resamples). These are descriptive companions to the held-out benchmark, not inferential tests; none of them contradicts the held-out conclusion. As one illustration, for math anxiety the standardized coefficient of the general score is .116 $[-.101, .322]$ and of the Anxious-specific residual $-.123 [-.288, .056]$. The complete table ships with the replication package.

F.6. Item-level benchmark: specification and complete results. Cell-level design: outcomes are the six item-level variables of Table F.4; the baseline design matrix contains 25 item-position indicators and the survey task-version covariate with its frozen missing-task indicator; the facial model adds the nine cell-level activation proportions standardized on training folds. Folds are participant-level (five-fold), 50 repetitions; binary outcomes are fitted by logistic regression and scored by mean negative Bernoulli log-likelihood (primary) with Brier score and area under the curve (AUC) as secondary descriptors; duration is fitted by linear regression and scored by root mean squared error (primary) with held-out R^2 change as secondary. The primary summary is the relative loss reduction $RLR = (\text{loss}_{\text{baseline}} - \text{loss}_{\text{facial}}) / \text{loss}_{\text{baseline}}$, positive when the facial model helps.

TABLE F.3. Person-Level Benchmark: Complete Held-Out Results
(100 Repetitions of Participant-Level Five-Fold Cross-Validation; Task-Corrected Primary Policy)

Outcome	Model	Median	2.5%–97.5% range	$P(\text{improved})$
First-attempt rate ($n = 169$)	Baseline R^2	−.0204	[−.0654, .0109]	—
	+ General	−.0075	[−.0319, .0062]	.17
	+ Specific	−.0089	[−.0390, .0102]	.29
	+ All nine	−.0624	[−.1537, −.0032]	.03
	+ Nuisance-adjusted	−.0069	[−.0363, .0073]	.20
Math anxiety ($n = 100$)	Baseline R^2	.1339	[.0527, .1891]	—
	+ General	−.0114	[−.0816, .0145]	.26
	+ Specific	−.0158	[−.0882, .0197]	.24
	+ All nine	−.1119	[−.2443, −.0139]	.01
	+ Nuisance-adjusted	−.0109	[−.0849, .0126]	.26
Mean log duration ($n = 169$)	Baseline R^2	.7711	[.7593, .7783]	—
	+ General	−.0059	[−.0122, −.0005]	.00
	+ Specific	−.0081	[−.0165, −.0013]	.01
	+ All nine	−.0264	[−.0388, −.0149]	.00
	+ Nuisance-adjusted	−.0030	[−.0086, .0005]	.06
Post-error help rate ($n = 100$)	Baseline R^2	−.0318	[−.1527, .0422]	—
	+ General	−.0214	[−.0739, −.0021]	.00
	+ Specific	−.0323	[−.0875, .0025]	.04
	+ All nine	−.1365	[−.2687, −.0417]	.00
	+ Nuisance-adjusted	−.0184	[−.0725, −.0015]	.00
Post-error skip rate ($n = 100$)	Baseline R^2	.6923	[.6535, .7113]	—
	+ General	−.0071	[−.0206, −.0011]	.00
	+ Specific	−.0100	[−.0264, .0012]	.05
	+ All nine	−.0402	[−.0804, .0020]	.05
	+ Nuisance-adjusted	−.0067	[−.0187, −.0005]	.00

Note. Rows after the baseline give increments ΔR^2 relative to the baseline of the same repetition. Ranges are 2.5th–97.5th percentiles across the 100 repetitions (empirical repeated-split stability ranges, not confidence intervals). $P(\text{improved})$ is the proportion of repetitions with $\Delta R^2 > 0$. The

two $n = 169$ outcomes use the corrected survey task covariate with the frozen missing-task indicator; the three $n = 100$ survey-outcome rows are numerically identical to the pre-correction release because their subsample and baselines contain no affected term. The all-nine model adds nine correlated predictors at the smallest $n = 100$ and is read as a demanding sensitivity model; predictor count and multicollinearity are acknowledged limitations of that model, not demonstrated causes. Negative medians are evidence of no improvement for the evaluated representations, not evidence of equivalence or harm.

TABLE F.4. Item-Level Benchmark: Complete Held-Out Results (50 Repetitions of Participant-Level Five-Fold Cross-Validation; Task-Corrected Primary Policy)

Outcome	Cells	Median RLR	2.5%–97.5% range	$P(\text{improved})$	Secondary Δ
First-attempt correctness	4,126	−.0042	[−.0093, −.0005]	.02	−.0032 (AUC)
Hint request	4,126	−.0092	[−.0185, −.0055]	.00	−.0066 (AUC)
Skip	4,126	−.0205	[−.0275, −.0116]	.00	−.0198 (AUC)
Log item duration	4,126	−.0073	[−.0120, −.0050]	.00	−.0102 (R^2)
Post-error help	1,722	−.0194	[−.0359, −.0095]	.00	−.0084 (AUC)
Post-error skip	1,722	−.0260	[−.0485, −.0180]	.00	−.0265 (AUC)

Note. $n = 169$ participants throughout under the primary missing-indicator policy; post-error outcomes use the 1,722 post-error cells. Under the complete-case sensitivity (162 persons; 3,952 ordinary and 1,645 post-error cells) the medians are −.0041, −.0082, −.0205, −.0074, −.0205, and −.0264 respectively; no conclusion changes. The benchmark evaluates concurrent whole-item activation aggregates only; event-locked and within-item dynamic representations were not tested, and the negative RLRs are not read causally.

Appendix G. External review, statistical quality assurance, claim contract, and reproducibility

G.1. External code-and-claims review. Before drafting, the complete analysis pipeline and its claims were subjected to an independent external review, which returned 30 findings. Final dispositions: 12 ACCEPT, 12 ACCEPT WITH RECALIBRATION, 5 REJECT (closed by documented validity decisions), and 1 NEEDS CHECK (closed after verification); all 30 are closed or explicitly deferred (two deferrals: a submission-time provenance audit and English-manuscript-stage wording checks, both scheduled rather than open). The review materially changed the reported analyses. It prompted (a) the self-report factor correction and full recomputation of the five-outcome benchmark in one versioned pipeline ([Section F.2](#)); (b) the all-nine and nuisance-adjusted sensitivity representations; (c) the effective-information reporting hierarchy (one-tick headline, AR(1) secondary, nominal reference) with corrected scale and scope language for the generalizability coefficients; (d) the sparse-support policy and recalibrated Joy language ([Section D.2](#)); (e) the mandatory provenance cross-walk wording ([Appendix B](#)); and (f) the machine-readable claim–evidence matrix and statistical-content quality assurance described next.

G.2. Second external review and the task-covariate correction. After both manuscripts in this research program were complete, an independent external pre-submission review audited the full project (manuscripts, code, contracts, frozen releases, and the first review’s closures). Its principal findings for this paper, with dispositions: (1) the criterion benchmarks’ task covariate implemented the historical collection label where the frozen contracts require the survey task version; all eleven benchmarks were rerun in a versioned correction release under a frozen missingness policy, with no conclusion direction changed ([Section F.3](#)); (2) the manuscript’s environment sentence overstated historical fit-runtime reproducibility; the reproducibility statement now distinguishes release consistency, the documented consumer environment snapshot, and the unrecorded fitting versions ([Section G.5](#)); (3) several results phrasings exceeded their calculations (effective-information magnitude, zero-shared-variance wording) and were bounded; (4) the outcome-blind, non-cross-fitted representation boundary and the theory-to-release supersession are now disclosed explicitly. This appendix summarizes the review findings and their dispositions; the full review report and a separate disposition record are not claimed as public package artifacts.

G.3. Statistical-content quality assurance. A dedicated quality-assurance release re-derives the registered manuscript statistics from the frozen analysis artifacts and runs 18 machine checks, summarized in the 15 rows of [Table G.1](#) (the manifest, configuration, and summary-reconstruction rows each cover two checks). The quality assurance itself is versioned: v1 was invalidated (a nonexistent-column reference produced an all-NA comparison that passed vacuously) and is retained only as an audit trail with an invalidation notice; v2 corrected the numerical checks; v3 added a claim-by-claim source trace; v4 (current) additionally verifies every traced source field against the literal file headers. The v3-to-v4 numeric cores are byte-identical. A deliberately corrupted fixture (one stale criterion cell) is included to confirm that the comparison machinery detects mismatches.

TABLE G.1. Quality-Assurance Machine Checks (All PASS)

Check	Evidence
Release manifests	Corrected analysis release: 20 payload files verified; measurement release: 26 payload files verified
Config bindings	Analysis configuration and reporting policy bound by SHA-256
Participant-cell accounting	179 test-log persons / 4,475 cells $\rightarrow$ 169 matched persons / 4,225 possible $\rightarrow$ 4,126 observed
Summary reconstruction	Criterion summary and delta tables re-derived from the repetition-level distributions; maximum absolute difference $< 10^{-15}$
Fold audit	2,500 fold rows; no train/test overlap
Design rank	12,500 fitted design matrices; 0 rank-deficient
Effective-tick derivation	AR(1) m_{eff} recomputed from lag-1 values; maximum absolute difference $< 5 \times 10^{-15}$
Release binding	Manuscript criterion table matches the release table cell-by-cell (maximum difference $< 10^{-15}$)
Mismatch detection	Intentional stale-cell fixture detected (difference 10^{-3} flagged)
Source immutability	6 of 6 consumed measurement-release artifacts unchanged since release
Self-report contract	0 = no (reference); 1 = yes; 2 = it depends
Nuisance roster	PC1-log-ticks Spearman $r = .3306$ reproduced
Sparse-support policy	Joy boundary 24 (494/500), Joy magnitude 20 (285/500) flagged as expected
G/D scope	9 channels $\times$ 3 scenarios with required role labels
Public identifiers	0 direct-identifier columns in any public artifact

G.4. Claim–evidence contract. Six core claims are bound to specific release artifacts in a machine-readable matrix; the manuscript makes no measurement or criterion claim outside them.

TABLE G.2. Claim–Evidence Matrix (Abbreviated; Machine-Readable Version in the Replication Package)

ID	Claim	Evidence	Scope
C1	Eight activation channels retain repeatable person ordering under the one-effective-tick stress test	$E[\rho^2] = .881\text{--}.963$	Primary eight excluding Joy; observed 25 item-position task
C2	Unequal 12/13 item splits reproduce activation ordering, with Joy as a lower-stability sparse counterexample	Raw $\rho = .542\text{--}.964$; adjusted $.703\text{--}.982$	All nine channels, activation
C3	The nine activation scores show descriptive cross-channel commonality without identifying a subjective-emotion factor	PC1 = 38.4%	Descriptive principal components and correlations
C4	The general score carries recording-volume information, but nuisance adjustment does not reverse the criterion result	PC1–log-ticks $r = .331$; maximum change .003	Three recording covariates
C5	The evaluated aggregate person representations do not improve the exploratory held-out benchmarks	Specific $\Delta R^2 - .032$ to $-.008$; all-nine $-.137$ to $-.026$	Five outcomes; evaluated representations only
C6	The evaluated concurrent whole-item representation does not reduce held-out predictive loss	0 of 6 outcomes with positive median RLR	Concurrent whole-item aggregates only

G.5. Reproducibility and environment. Reproducibility is stated at three levels. The frozen releases are internally consistent and re-derivable; the consumer and quality-assurance environment that reads them is documented by `DESCRIPTION` plus the shipped `sessioninfo.txt` snapshot; and the R and package versions of the original fitting runtime were not recorded at fit time and are documented as unrecorded in the frozen environment contract. The package contains no `renv` lockfile, and neither an exact clean-environment restore nor historical fit-version recovery is claimed as a demonstrated property. The frozen releases consumed by this manuscript, with their integrity hashes, are listed in [Table G.3](#).

The replication package [TO BE ADDED: repository URL] contains selected read-only source modules for the reported evidence chain, disclosure-screened aggregate results, verification gates, the machine-readable claim–evidence matrix, and a technical report that develops the methodology from its premises. Input-bound configuration and superseded branches are documented as excluded. The governing disclosure rule is that no released object may contain, or permit reconstruction of, a value at

TABLE G.3. Frozen Release Artifacts and SHA-256 Integrity Hashes

Artifact	SHA-256
Corrected analysis release (Phase 9C)	dd9574268d0d7eaaa06c3363796e6dd8a43db1d1715c2678972fe46a05f23945
Statistical QA v4 (success)	c9dcd5a116dc6ca1126f4acb6049b4ea8e571add2d93d16617d122dbeca56985
Statistical QA v4 (manifest)	62b21b109c1a32fd43b1e7e88e14323791d93b4c6344b11185e339e518598350
Manuscript freeze v3	dcd02a08157a05f5cd67d51a0457f04cd2456d527ca195b7e384cc4c04d85962
Pre-draft audit v3 (success)	dcf14212afef88ff53e2ae16e46623b09be753adb9bec80a18bdbd4abf4d3c1c
Pre-draft audit v3 (manifest)	3f2565045ba68c5db31c1910a80c7e0ace771777811f59670148e2d2431d0dbc
Phase 9D release (success)	6f1e1d16c1fc875193da739b2b2ba21d52100271db4a8e7e162d7cf58226bedd
Phase 9D release (manifest)	cc162ad102876b8a9b531adf780fec66f50338f84a0d469f8c10d0c81726e530

the grain of a participant or of a participant-by-item cell. The disclosure scanner is an explicit stage of the full verification suite and release review; it is not invoked by every runnable entry point.

The package’s 35-target manuscript contract classifies artifacts by what can be verified publicly. The runnable exhibit layer rebuilds ten table files and four public figures from aggregates. Other targets are identified as verified-static transcriptions, public-number verification targets, partially withheld artifacts, or withheld artifacts. In particular, [Figure 1](#) and panel (d) of [Figure 5](#) are withheld because they require participant-grain material; readable generators or aggregate substitutes document what can be checked without those rows. The excluded configuration includes restricted paths, content hashes, and participant-specific alignment information. The technical report states the public alignment, weighting, missingness, and analysis constants supported by released authorities, but does not claim to reconstruct every input-bound configuration value.

G.6. Administration-mode descriptives: derivation and verification. The administration-mode facts reported in this article (the 42/127 split, the wave dates, the mode composition of the 99 unobserved cells, the per-mode tick statistics, and the mode descriptives of [Table E.4](#)) postdate the frozen releases: they were established during the present revision, after the archival record was found to contradict the online-only administration description. They are derived by a

versioned script in the manuscript repository (`figures/src/00_common.R`) that reads only the immutable archival stores (the survey table’s mode variable, the facial-database session timestamps, and the cell-coverage store) together with the restricted release person table, re-derives the registered mode quantities at build time, asserts their anchors before any figure is written, and writes four machine-readable authorities under `data/`: `mode-descriptives.csv`, `mode-dates.csv`, `channel-descriptives.csv`, and `mode-by-channel.csv`. The manuscript’s number-verification gate (`tools/verify-numbers.py`) checks the registered mode facts in text and tables against those authorities, using the same mechanism applied to the registered frozen-release quantities. The mode quantities are archival descriptives outside the claim matrix of [Table G.2](#); no claim of the matrix depends on them, and they license no effect statement.